“A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA”



Dissertation submitted to

Dr. N.T.R. UNIVERSITY OF HEALTH SCIENCES,

Vijayawada

In partial fulfilment for award of degree of

AYURVEDA DHANVANTARI

Master of Surgery (Ayurveda)

In Prasuti Tantra & Stree Roga

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DECLARATION

I hereby declare that this dissertation entitled "A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA" is a bonafide and genuine research work carried out by me under the guidance of Dr. KOLA VIJAYA KUMARI, M.D (Ayu) Professor & H.O.D Department of Prasuti Tantra & Stree Roga, Dr. N.R.S Govt. Ayurvedic College, Vijayawada,during the period 2022-25.

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ACKNOWLEDGEMENTS

I humbly begin by offering my deepest gratitude and my Pranam to **LORD DHANVANTARI**, the divine physician, whose blessings have guided me in the successful completion of this work.

I would like to extend my deep gratitude and courtesy to my guide **Dr. Kola Vijaya kumari, M.D (Ayu), Professor and Head of the Department of Prasuti Tantra and Stree Roga** at Dr.N.R.S Government Ayurvedic college , Vijayawada. Her Valuable suggestions, close involvement with this work, keenness, and enthusiasm were instrumental in achieving this milestone. It has been a great privilege and honour to work and study under her guidance.

I am very much thankful to **Dr. R. Vijaya Santhakumari, M.D (Ayu)**, P.G Diploma (YOGA) Associate Professor, Prasuti tantra and Stree roga, Dr. N.R.S Government Ayurvedic College, Vijayawada, for her consistent encouragement and for sharing her knowledge which enlightened me a lot.

I am very much thankful to **Dr. J. Mahitha, Dr. B. Ratna Manikyam, Dr. D. Mouli Naik** Assistant Professors of Prasuti tantra and Stree roga Department, Dr.NRS Government Ayurvedic College, Vijayawada for their valuable suggestions.

I convey my sincere thanks to **Dr. P. Sai Sudhakar, M.D (Ayu), Ph.D,** Principal of Dr. N.R.S Government Ayurvedic College, Vijayawada for providing facilities for the research work.

I convey my sincere thanks to **Dr. P. Raja Rajeswari, B.A.M.S,** Retd. Superintendent, A.L Government Ayurvedic Hospital, Vijayawada, for providing necessary facilities during my study.

I am thankful to Rasa sastra and Bhaishajya kalpana department for their support during drug preparation. Special thanks to **Dr. K Srihari M.D (Ayu),** Associate Professor, Head of the department of Rasa Sastra and Bhaisajya Kalpana.

I am thankful **Dr. B. Raja Kamalakar, MD(Ayu),** Assistant Professors Department of Panchakarma, **Dr. B. Prasanna,** Assistant Professor, Department of Dravyaguna for teaching me Statistics and research methodology.

I am very thankful to the staff of library **Mr. Deekshitulu** for his kind co-operation which was very much necessary for the successful completion of this work.

I am very thankful to my Parents **Dr. Sanjay Patel, MSc, PhD (Statistics)** and **Mrs. Nikita Patel** and my brother **Harshal Patel**, who supported and encouraged me throughout my study.

I am thankful to **Mr. B. Sreeram Madhav, MSc Statistics** from Badruka college, Osmania university, for helping me in statistical analysis.

My special thanks to all my seniors, my batch mates, juniors and all other college & hospital staff who helped me in different times for the successful completion of this work.

I also thank my patients who rendered help directly or indirectly in successful completion of this project without whom, my research work would have been just a futile exercise.

I thank them with profound indebtedness for their faith and confidence in me.

Dr. PATEL NIDHI SANJAYBHAI

INDEX

SL. NO	CONTENTS	PAGE. NO
1.	INTRODUCTION	1-3
2.	AIMS AND OBJECTIVES OF THE STUDY	4
3.	REVIEW OF LITERATURE a) AYURVEDIC REVIEW b) MODERN REVIEW c) DRUG REVIEW	5-42 5-19 20-29 30-42
4.	MATERIALS AND METHODS	43-54
5.	OBSERVATIONS AND RESULTS	55-74
6.	DISCUSSION	75-81
7.	CONCLUSION AND SUMMARY	82-85
8.	REFERENCES	86-91
9.	LIST OF TABLES	92
10.	LIST OF FIGURES	93
11.	LIST OF GRAPHS	94
12.	ANNEXURES	95-105

ABBREVIATIONS

1.	A.H.Su	-	Ashtanga Hridhayam Sutra sthana
2.	A.H.Sha	-	Ashtanga Hridhayam Sharira sthana
3.	A.H.U	-	Ashtanga Hridhayam Uttara sthana
4.	A.S.Su	-	Ashtanga Sangraha Sutra sthana
5.	A.S.Sha	-	Ashtanga Sangraha Sharira sthana
6.	A.S.U	-	Ashtanga Sangraha Uttara sthana
7.	AT	-	After Treatment
8.	Bha.Pra.Pu	-	Bhavaprakasha Purvakhanda
9.	Bhe.Sa	-	Bhela Samhita
10.	Bhe. Sha.	-	Bhela Samhita Sharira sthana
11.	Bha.Rat	-	Bhaisajya Ratnavali
12.	BT	-	Before Treatment
13.	Ch.Su	-	Charaka Samhita Sutra sthana
14.	Ch.Sha	-	Charaka Samhita Sharira sthana
15.	Ch.Chi	-	Charaka Samhita Chikitsa sthana
16.	FSH	-	Follicle Stimulating Hormone
17.	GnRH	-	Gonadotrophin Releasing Hormone
18.	Ha.Sa.	-	Haritha Samhita
19.	HBsAg	-	Hepatitis B surface antigen
20.	Hb	-	Haemoglobin
21.	HCV	-	Hepatitis C virus
22.	HIV	-	Human immune deficiency virus.
23.	HPO	-	Hypothalamo Pituitary Ovarian
24.	HTN	-	Hypertension
25.	IPD	-	In Patient Department
26.	Ka.Sa.	-	Kashyapa Samhita
27.	Ka. Khi.	-	Kashyapa Khila sthana
28.	LH	-	Luteinising Hormone
29.	OPD	-	Out Patient Department
30.	P	-	Probability Value
31.	R.T	-	Rasa Tarangini
32.	SD	-	Standard Deviation

33.	SE	-	Standard Error
34.	Sha.Sa.Pu.	-	Sarangadhara Samhita Purvakhanda
35.	Su.Su	-	Sushruta Samhita Sutra sthana
36.	Su.Sha.	-	Sushruta Samhita Sharira sthana
37.	Su.Ni	-	Sushruta Samhita Nidana sthana
38.	Su.Chi.	-	Sushruta Samhita Chikitsa sthana
39.	USG	-	Ultrasonography
40.	Y. R	-	Yogaratnakara

INTRODUCTION

INTRODUCTION

In the present era many changes are seen in the life of women as compared to previous. The present era being a challenge and competitive to the woman where in woman has entered all the fields which were with risk. To achieve the goals of such scenario, women health is the prior factor to consider & to maintain its equilibrium.

Ayurveda the science of life the treatise where the calculative parameters for physicals, psychological, social and spiritual health are focusing in gives utmost importance and focusses on the woman health as it is believed that health of family, society & nation largely depends on the health of its women.

The core definitions of Stree in Ayurveda denote it emphasise on woman health.

Acharya Charaka postulates

"प्रजननार्थम् स्त्रियाः सृष्ट्याः ||"

i.e., the process of procreation of the universe is carried forward by the woman.

Therefore, the person who carries it is defined as Stree. The healthy procreation is attained through healthy menstruation which is having regulation of hormones in all the days of her life. In woman during reproductive age an essential phenomenon is menstruation which invokes periodical shedding of endometrium which provide a bed for fertilized gamete, the process when fails results in physiological menstruation. This phenomenon is called as "masanumasika rajah pravritti" or "artava pravritti". There are many pathologies pertaining to artava pravritti viz kashtartava, anartava, artavriddhi, artavakshaya etc.

Artava is expelled out of uterus through vagina visually manifested for 3-5 days in a biological rhythm throughout reproductive period of woman (from menarche to menopause).

The present study deals with characters of artava quantitatively and pathologies pertaining to it which causes discomfort to the woman.

Acharya Charaka has mentioned,

नैवाति बहु नात्यल्पम् आर्तवं शुद्धिमादिशेत् । (च. चि.30/224)[1]

The menstrual blood of its own physiological quantity is considered fruitful.

(4 anjali) Any deviation from that quantity is understood pathological and incapable of serving purpose.

The functionality is also calculated through its characters like interval between two successive menstrual cycles, the duration of menstrual cycle and not associating with any pain.

मासात् निष्पिच्छ दाहार्ति पन्चरात्रानुबन्धि च । (च. चि.30/224)[1]

The normal menstruation is that which has interval of 28-35 days resembling Gregorian calendar to Lunar calender, 5 days of duration & not associated with pain and burning sensation. It should neither be scanty in terms of quality or quantity nor should be excessively flowed.

In Ayurveda, it is clearly mentioned that the artava which is scanty – ksheenartava doesn't allow the proper functioning of ritucharya and thereby it affects the reproductive health of the woman.

Acharya Sushruta had explained artava kshaya which is characterised by yathochita kala adarshana, alpata and yoni vedana, which could be compared to hypomenorrhea based on their signs & symptoms. Menstruation occurring more than 35 days apart (oligomenorrhea) is unduly scanty lasting for less than 2 days along with increased intermenstrual gap is hypomenorrhea.

As Artavakshaya is explained as precursor of nashtartava according to acharya Sushruta, the involvement of dosha mainly vata causing 'sanga' type of sroto dushti. Here, the main dosha is vata (mainly apanavata) which plays a major role in physiology & pathology of reproductive tract and basti considered to be the best treatment for vataja disorders.

Thus, the present study was undertaken to address the hypomenorrhic patients with oral medicine & locally uttarabasti.

Uttarabasti is the intrauterine application of drug helps in rejuvenation of endometrium, correction of hyperplasia (vice versa), removal of lumen blockages thereby restoring the normal functioning of H-P-O axis by absorption of active principle of medicine locally.

For the treatment of artavakshaya, acharya sushruta has postulated the line of management as shodhana and administration of agneya Dravya.

Thus, the composition satisfying the entire parameters of treatment i.e., Lashuna Kalpa orally & Rasnadi Taila uttarbasti explained in Ashtanga sangraha.

The drug lashuna is having property of artavajanan, raktapitta janana, Vedana stapana and also having action of uterine stimulant. Rasnadi taila explained in ashtanga sangraha is quated as,

……प्रणीतो अनिलार्तिहरः ।

(A.San.U.39/36)[58]

So, here the ingredients of Rasnadi taila having the property of Vatahara and Lekhana action (Scraping action).

The efficacy of drug along with its route of administration is studied through the post treatment quantitative statistical study in a sample of 30 size in present study.

AIMS & OBJECTIVES

AIMS AND OBJECTIVES OF STUDY

This study entitled

'A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA'

is aimed to fulfil the following criteria.

- To maintain physiological health of reproductive woman.
- To establish easily available non-controversial drug Lashuna kalpa in Hypomenorrhea as per ayurvedic classics.
- To study therapeutic efficacy of Rasnadi tailam Uttar basti in Artava kshaya w. s. r. to Hypomenorrhea.

NEED OF STUDY: -

- Different phases of menstrual cycle play a great role in women's health.
- So, To care woman's menstrual health unto avoid delayed menstrual complications.
- In allied science, It is treated with hormonal & other drugs which produce massive side effect so to have simple herbal management.
- Ayurvedic drugs have the properties of preventing systemic disorders to bring back the vitiated doshas into their normal state and shows the normal menstrual cycle.
- Thus, to prevent unwanted side effects by using modern medicine the present study is aimed to establish the efficacy and safety of Ayurvedic formulation towards female health.

REVIEW OF LITERATURE

REVIEW OF LITERATURE

Ayurvedic Disease Review

To understand appropriate etiopathogenesis and treatment of any gynaecological disorder, knowledge of STREE SHAREERA (anatomy & physiology of female reproductive system) is necessary.

SHRONI:

The measurement of chest of a male and waist of female is identical. The measurement of waist of female is twenty-four angula's.

श्रोणिरत्रोरुसन्धस्तात्स्मरमन्दिरोपरितन्दविभागः | (S.Su. 35/12, Dalhana)

Shroni is the area between the urusandhi and the smaramandira. Urusandhi refers to joint between the head of femur and pelvic bones.[2]

चतुर्विंशत्यङ्गुलविस्तारमुक्तत्प्रमाणास्त्री श्रोणिझेया | (S.Su. 35/12, Dalhana)

The chest in male and waist in the female measures 24 angulas.[3]

SHRONI ASTHI:

द्वेश्रोणिफलकेएकंभगास्थि | (C. Sha.7/6)

श्रोण्यांपञ्च, तेषांगुदभगनितम्बेषुचत्वारि, त्रिकसंश्रितमेकम् | (S. Sha.5/19)

Table No: 1 Bones of Shroni

According to Charaka[4]	According to Sushruta[5]
3 Bones	5 bones
2 Shroni phalakas 1 Bhagasthi	1 Gudasthi 1 Bhagasthi 2 Nitambasthi 1 Trikasthi

SHRONI SANDHI:

त्रयीकटीकपालेषु | (S. Sha. 5/26)[6]

गुदभगनितम्बेषुसामुद्राः, कटिकपालेषुतुन्नसेवन्यः | (S. Sha.5/27)[7]

Flat bone of Pelvis has three joints which are tunna sevani type. In Bhaga and Nitamba the bone has samudga type of joints.

BHAGA (Vulva):

द्वादशांगुलानिभगविस्तारः | (S. Su. 35/12)[8]

The shape of the bhaga is compared to the triangular-shaped leaf of pippali. Bhaga measures 12 angulas.

ASHAYA:

There are seven ashayas in human body, which are common for both male and female.

स्त्रीणांगर्भाशयोरष्टमइतिः | (S. Sha. 5/43)[9]

Apart from the sapta ashayas or the seven ashayas, a female body has 8th ashaya, the garbhashaya.

तस्यास्तृतीयेत्वावर्तेगर्भशय्याप्रतिष्ठता | (S. Sha. 5/43)[10]

पित्तपक्वाशयोर्मध्येगर्भशय्यायत्रगर्भस्तिष्ठति | (S. Sha. 5/39)[11]

Garbhashaya is situated in the third avarta of the yoni. It is situated behind the urinary bladder, between the pittashaya and pakwashaya.

यथारोहितमत्स्यस्यमुखंभवतिरुपतः । (S. Sha.5/44)[12]

The shape is compared to that of a rohitha matsya mukha. The garbhashaya is hollow inside like the mouth of a fish.

YONI:

शंखनाभ्याकृतिर्योनिस्त्यावर्तीसाप्रकीर्तिता ।

तस्यास्तृतीयेत्वावर्तेगर्भशय्याप्रतिष्ठिता || (S. Sha. 5/43)[13]

The shape of the yoni is compared to the shape of a shankha nabhi or hallow portion of the conch shell. It has three avartas or circular folds. Garbhashaya is situated in the third avarta or uppermost fold.

STREE PESHI: Women possess 20 extra Peshis.[14]

- Ten peshis in STANA – five in each Stana. They develop in adolescence.
- Four peshis in APATYAPATHA – two spreading inside and two situated in orifice in circular fashion.

- Three peshis in GARBHASHAYA CHHIDRA OR GARBHAMARGA.
- Three are responsible for entry of SHUKRA (semen) and ARTAVA (ovum).

STROTASA:[15]

There are two srotas in connection with the special organs of female.

आर्तववहेद्वे | तयोर्मूलंगर्भाशय आर्तववाहिन्यश्चधमन्यः | (Su. Sha. 9/5)

These carry the artava and are called the artavavaha srotas. Origin of the two artavavaha srotas are the garbhashaya and the artava vahi Damani's.

तत्रविध्दायांवन्ध्यात्वम्मैथुनासहिष्णुत्वमार्तवनाशश्च | (S. Sha. 9/12)

Injury to these strotas leads to infertility, dyspareunia, and amenorrhea.

STREE SHARIRA KRIYA:

The knowledge of the physiology of the female reproductive system has its own role in the diagnosis of menstrual abnormalities.

ARTAVA:

❖ Definition:

ऋतौ भवमार्तवंस्त्रीणां यदपत्यमार्गात् शुद्ध ईषत् कृष्णं विगन्धं च ।
वायु प्रेरितम् लोहितं प्रवर्तते तदार्तवमुच्यते || (अ.हु.शा. 1/1)[16]

The substance which flows out of the apatyamarga having red color with mild black discoloration and without any gandha due to the action of vayu in a cyclic manner or at a specific period is called "artava".

स्त्रीणां गर्भोपयोगी स्यात् आर्तवं सर्व सम्मतम् । तसामपि बलं वर्ण शुक्रपुष्टिं करोति हि ||
एतेन स्त्रीणां सप्तमो धातुरर्तवं शुक्रमष्टममिति बोधितम् ।आशयाधिक्यत्वात्……..||
(भा. प्र.पू.3/188)[17]

Acharya Bhavamisra has explained that the women possess one extra dhatu just as presence of an extra ashaya (garbhashaya), thus they have artava as seventh dhatu and sukra as eighth dhatu. All the functions related to conception are performed by artava and functions like bala, varna and Pushti are promoted by sukra.

❖ Formation of Raja / Artava:

रसादेव स्त्रिया रक्तं रजः संज्ञा प्रवर्तते || (सु. सू. 14/6)[18]

तथा रक्तमेव च स्त्रीणां मासे मासे गर्भकोष्ठ अनुप्राप्य त्र्यहं प्रवर्तमानं आर्तवमित्याहुः || (अ.सं.शा.1/9)[19]

Acc to Ayurveda, the rajas or artava is formed as an upadhathu from rasa dhatu i.e., artava is formed from sara bhaga of rasa dhatu. This rakta or raja reaching garbhashaya and flows out of it periodically for 3 days is considered the artava.

❖ Artava nirmana kala:

एवं मासेन रसःशुक्रं स्त्रीणां च आर्तवं भवति || (सु.सू. 14/14)[20]

It takes one month for transformation of rasa dhatu into proper artava or rajas.

❖ Artava swaropa:

आर्तवं शोणितं तु आग्नेयम्…… | (सु.सू. 14/7)[21]

स्त्रीणां रक्तलक्षणम् आर्तवं गर्मकृत् च || (सु.सू. 15/9)[22]

Artava is agneya in nature and bears the characteristics of rakta, though dominant mahabhoota is agni, the other mahabhoota also reside in it as sookshma form.

❖ Artava pramana:

चत्वारो रजसः स्त्रियाः | (अ.हु.शा.3/81)[23]

स्त्रीणां रजसो अन्जलयः चत्वारः || (अ.स.शा.5/93)[24]

The pramana of artava in a female is 4 Anjali.

❖ Artava pravritti kala:

तद् वर्षात् द्वादशात् ऊर्ध्वं याति पन्चाशतः क्षयम् || (सु.सू. 14/6)[25]

According to brihattrayee, in females artava darshana starts at the age of 12 years and it ceases at the age of 50 years. According to Acharya Kashyapa, age of menarche is 16 years, but it may vary according to the ahara, vihara of that girl.

❖ Aratava utpatti hetu:

According to karya karana vada, every karya has a karana, when we look into different opinion of Acharyas, the artava utpatti is clearly found with its hetu includes the following

- Kala
- Swabhava
- Dhatu paripurnatva
- Karma
- Vayu

❖ Suddha artava lakshana:

मासात् निष्पिच्छ दाहार्ति पन्चरत्रानुबन्धि च । नैवाति बहु नात्यल्पम् आर्तवम् शुद्धमादिशेत् ||
गुन्जाफल सवर्ण च पद्मालक्तक सन्निभम् । इन्द्रगोपक सन्काशं आर्तवं शुद्धिमादिशेत् ||
(च.चि.30/225)[1]

According to Acharya Charaka, the normal menstruation is that which has

- Intermenstrual period of one month
- Duration of blood flow for 5 days
- Not associated with pain or burning sensation
- The menstruated blood devoid of Pichila guna
- Neither **very scanty, nor very excessive** in amount.
- The color of menstruation discharge resembling red juice of lac, red lotus flower. Fruit of gunja or like indragopa.

शशासृक्प्रतिमंयत्तुयद्वलाक्षारसोपमम् |
तदार्तवंप्रशंसन्तियद्वासेनविरञ्चयेत् || (Su. Sha.2/17)[26]

According to Acharya Sushruta, the shuddha artava should be

- Red in color like blood of rabbit or like laksha rasa or ishat Krishna varna.
- It should not stain the clothes

Acharya vagbhata has explained same features of shuddha artava along with the added lakshana “vigandha” which may be due to dushita pitta.

Table No 2: Lakshana of Raja Srava

S.No.	Acharya opined	Duration mentioned	Color	Amount of blood loss	Character staining	Associated symptoms
1	Charaka	Pancharatra (5 D & 5 n)	Gunja phala, padma alaktaka color, indragopa coloured	Na ati bahu Na atyalpam	-	Not associated with daha, arti pichilatva
2	Sushruta	-	Laksha rasa, shasasruk	4 anajli	Yat vaso na ranjayet	-
3	Ashtanga Hridhaya[27]	3 days	Laksha rasa Shasasrik	4 anajli	Dhautam yat cha virajyate	-
4	Ashtanga Sangraha[28]	3 days	Sasha rudhira, laksha rasa	-	Dhautam cha Virajyamanm	-
5	Hareeta[29]	7 days	-	-	-	-

❖ Rutuchakra

The duration of menstrual cycle is 1 month (chandramasa-28 days) divided into 3 phases.

1. रज स्राव काल – Menstrual phase – 3 to 5 days
2. ऋतुकाल – Proliferative phase and Ovulation
3. ऋतुव्यतीत काल – Secretory phase / Postovulatory phase

1) Rajasrava kala

The Rajahsrava kaala or the menstrual phase comes at an interval of one month.

मासेनोपचितम् काले धमनीभ्यां तदार्तवम् ।
ईषत् कृष्णं विवर्णं च वायुर्योनिमुखं नयेत् ॥ (Su.Sha.3/10)[30]

Explaining about the physiology of this cycle, Acharya Sushruta has said that the dark coloured blood gets collected during the whole month by the Artava vimochini dhamani in the Garbhashaya and this collected blood gets expelled through the Yonimukha monthly by the action of Vayu (Apanavayu). The character of the blood expelled during this kaala has also been mentioned.

2) Rutukala

Rutukaala is that phase of Rutuchakra which comes after the Rajahsrava kaala.

स्त्रीणां रजोदर्शनात् षोडशनिशाः तत्र भवमार्तवम् ॥ (Ba. Pa. Purva.3/206)[31]

Rutukala is on 16th day of menses when ovulation occurs.

This kala is apt time for conception. The reason for conception during Rutu kaala has been explained very well as follows – As the Lotus flower closes itself after sunset, similarly the Yoni of women gets constricted after Rutu kaala and does not accept Shukra or the entry of Beeja (sperms) into its inner components i.e. uterus, fallopian tubes etc.

Duration

ऋतस्तु द्वादशरात्रं भवति दृष्टार्तवः। अदृष्टार्तवाऽप्यत्येके भाषन्ते ॥ (Su.Sa.3/6)[32]

The duration of this phase is twelve days according to Acharya Sushruta.

Acharya Vagbhata considers this duration as twelve days or sixteen days or it may also be of one month.[33]

Dalhana clarifies that while taking 16 days into consideration first 3 days and last 1 day excluded due to yoni sankocha.[34]

Then it is 12 days.

3) Rutuvyatita kala

ऋतौव्यतीते नार्यस्तु योनिः संव्रियते तथा । (Su. Sa.3/9)[35]

This is the phase of Rutu chakra just after the Rutu kaala and it ends with the onset of the Rajahsrava kaala.

This phase has been mentioned in very short as the phase of the Rutuchakra when there occurs the closure or constriction of the Yoni.

Because of this closure, entry of sperms is not allowed so no conception in this phase.

Table No 3: Phase of menstruation and predominance of dosha

Phase	Dosha dominance
रज स्राव काल	Vata
ऋतुकाल	Kapha
ऋतुव्यतीत काल	Pitta

DISEASE REVIEW

HISTORICAL REVIEW

Vedic era:

It is nowhere clearly mentioned in vedas regarding artavakshaya. There are scattered references in Rigveda regarding treatment of anartava (also like treat artava kshaya). In a book called atharvavediya karmaja vyadhi nirodha, it has been mentioned the etiopathogenesis of nyunartava and its management is also mentioned. In vandhya kalpa druma by R.J.Sharma there is mentioning of "nyunartava" as artava kshaya.

Samhita kala: Sushruta Samhita

Sushruta Samhita mentions the lakshanas and chikitsa of artava kshaya in Sushruta Samhita sutra sthana chapter 15. While describing the artava dosha, he has mentioned ksheenartava one of the symptoms of artava kshaya in shareera sthana. In shareera sthana, Sushruta explained the term artavanasha caused due to avarana of dosha and claims if it is nashtartava, it lands in to vandhya.[36]

Charaka Samhita:

Acc to Acharya Charaka, the Lakshana of shuddha artava itself is denoted through quantity of it as one of its parameters for optimum i.e. to be shuddha artava, it should not be less in quantity. The artava being extremely low is coined as term "arajaska" whereas the artava which is incapable of producing an offspring due to its lacuna in quantity is termed ksheena artava dushti.[37]

Ashtanga Sangraha & Ashtanga Hridaya:

Acc to Vagbhata Acharya, artava kshaya is not a special entity of pathology but he considers the word "ksheenartava" and uses it as synonyms with artavakshaya. Explained it as a symptom of lohitakshaya yonivyapad.[38]

Kashyapa Samhita:

Acharya Kashyapa terms artavakshaya as “alpapushpa” which leads to vandhyatva means anovulatory cycle- to maintain normal menstruation healthy ovulation is needed. He mentioned it as a vyadhi which is cursed by anuvasana vasti.[39]

Bhela Samhita:

He opines the artavakshaya as alpartava.[40]

Sarangadhara Samhita:

Acharya Sarangadhara mentions a pathological entity “ksheenartava dushti” a krichra sadya artava dushti and mentioned the management as per the involvement of dosha.[41]

Bhavaprakasha Samhita:

Acharya Bhava Mishra opines like Sushruta Acharya. He explains ksheenartava as artava kshaya & explains in detail with nidana etc.[42]

ARATAVA KSHAYA

In Ayurveda all the classical texts explained menstrual disorders under same entity “artava dushti”. A very meagre or less information is available regarding rajonasha, nashtartava, artava kshaya and artava ativridhi.

The term artava kshaya or ksheenartava sounds synonymous but explained as different entities in Samhitas.

The term artava kshaya is combination of 2 words ‘artava’ + ‘kshaya’.

The word artava denotes that which occurs cyclically and kshaya from kshi dhatu which means to leave or to get reduced.

The meaning of kshaya is postulated as

1. Hrasa, nyuna – Charaka Samhita
2. Nyunata, apachaya by vishala shabdakosa sagar.

The author venimadhava sastry has explained artava kshaya in his book Ayurvediya Maha Shabdakosha as

क्षीणार्तवं नाम आर्तवस्य स्वप्रमाण हानि ||[43]

That means the reduced quantity of artava from its normal pramana.

As a symptom artavakshaya could be observed in following yonivyapad

1. Arajaska
2. Vataja yonivyapad

- Artavakshaya is defined as,

  According to Acharya Sushruta,

  आर्तवक्षये यथोचितकालादर्शनमल्पता वा योनिवेदना च । (Su.su.15/12)[36]

- In the event of deficiency or loss of Artava, the menstruation does not appear in it's appropriate time or is delayed (Intermenstrual period is prolonged), is **scanty,** yoni vedana (abdominal pain) and does not last for 3 days.
- Acharya Bhela described that even though the blood circulates throughout the body, it cannot nourish the artava due to vikrithi and alpata of rakta dhatu which in turn leads to artava kshaya.[40]

Nidana

- The causative factors explained in classics can be divided into for the sake of convenience, viz. Samanya (general) and Vishesha (Specific).

Samanya Nidana

In classics, Artava is considered as Updhatu of Rasa Dhatu. The factors responsible for Rasa Dhatu Kshaya are responsible for Artava Kshaya. According to Sushruta Rasa Kshaya is one of the causative factors of Dhatukshaya.

Sushruta described Mithya Ahara Vihara as causative factor for Dhatukshaya. It is Samanya Nidana of Artava Kshaya.[44]

- **Vishesha Nidana**

The factors responsible for Artava Utpatti if vitiated by any cause lead to Artava Kshaya. These factors are taken as Vishesha Nidana of Artava Kshaya. The Vitiated Artavotpadaka Ahara, Agni (Jathragni, Rasagni), Rasadhatu and Artavavaha Srotas are main factors for Artava vitiation. Artava is Pittapradhana Agneya in nature. So opposite Vata Kaphavardhaka Ahara Vihara are Vishesha Nidana of Artava Kshaya.

Sushruta has clearly mentioned that, main causative factor that come into action in the production of Artava Kshaya are Vata and Kapha Dosha in combined form.[45]

Table No 4: Nidana of Artavakshaya

Aharaja	Viharaja	Manasika
Anashana	Ati samshodana	Shoka
Asatmya ahara sevana	Ati samshamana	Bhaya
Teekshna ahara sevana	Ati vyavaya	Chinta
Abhojana	Atapa sevana	Krodha
Atibhojana	Vega dharana	Mana santapa
Atyalpa Bhojana	Ratri Jagarana	Irshya
	Divaswapna, Alasya	

Purva Rupa:

- Purva Rupa of Artava Kshaya is not described in classics. In Bhavprakasha specific desire of the women suffering from Artava Kshaya have been enlisted i.e. she desire Katu, Amla, Lavana, Ushna, Vidahi, Guru Ahara and Phala Shaka and beverages etc.[46]

Katu etc. Ahara increase Artava thus, the desire to consume these substances is probably due to deficiency of Artava. Hence it can be considered as Purvarupa of Artava Kshaya.

Rupa:

Sushruta explained the symptoms of Artava Kshaya as[36]

1. **Yathochita Kaala Adarshanam** -

- Delayed menarche- Occurring at the age of 15 years or later can be considered as Artava Kshaya.
- Prolonged intermenstrual period.
- Duration of menstruation less than normal duration i.e. less than 3 days.

2. **Alpartavam –**

- According to Shabdakalpadruma “Alpa” means “Kshudra Pramana.” If menstrual blood flow is in less quantity i.e. less than 4 Anjali or less than 35 ml it should be considered as Artava Kshaya.

3. **Yoni Vedana -**

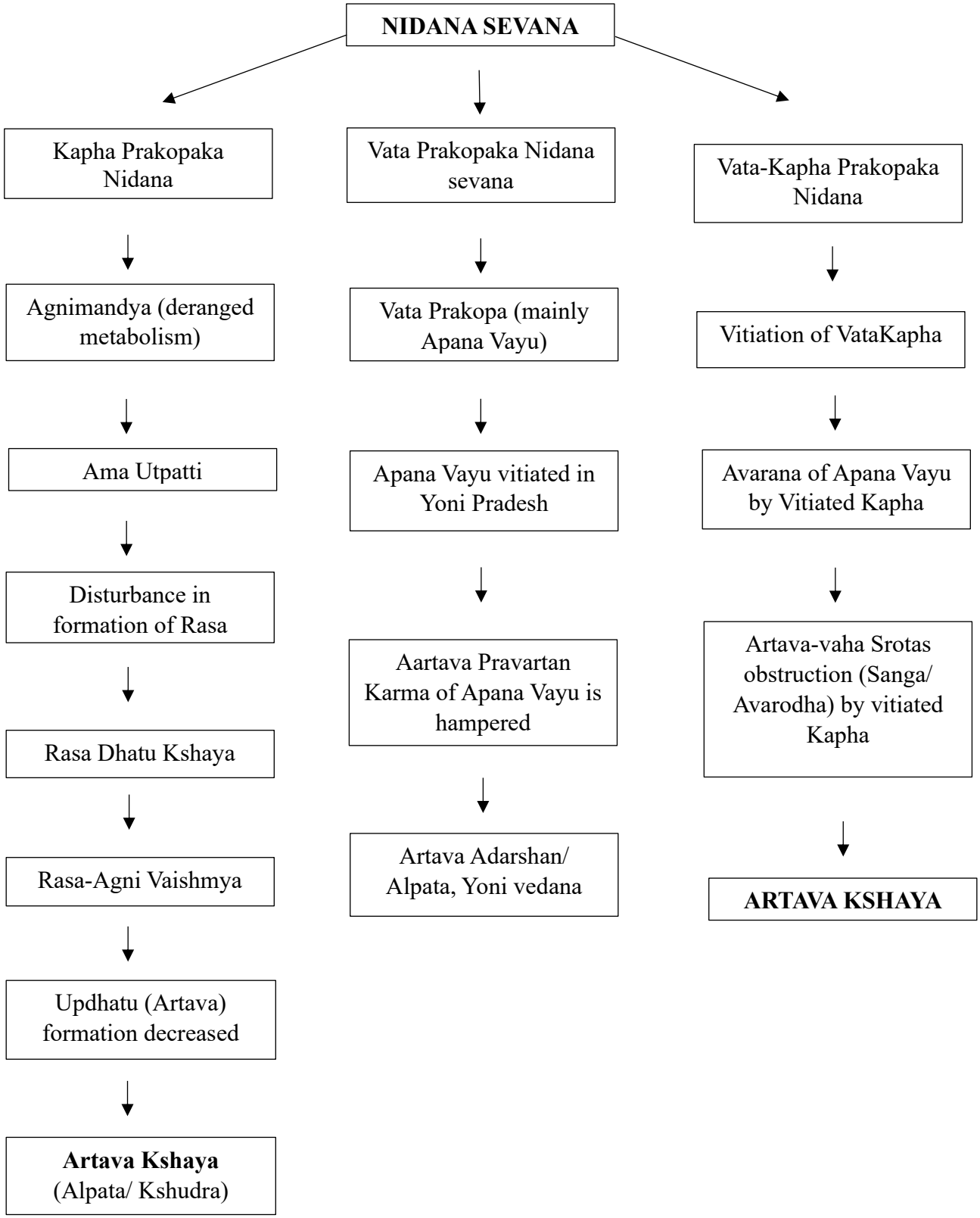
- Due to Artava Kshaya, vitiation of Vata mainly Apana and Vyana Vata causes Yoni Vedana. Yoni Vedana can be considered as spasmodic, radiating, and infrequent pain during menstruation.

Samprapti ghataka:

Table No 5: Samprapti Ghataka

Dosha	Avarita –Vata (Apana and Vyana), Avaraka –Kapha
Dushya	Rasa, Rakta
Upadhatu	Artava
Agni	Jathragnimandhya; Dhatvagnimandhya
Srotas	Rasavaha, Artavavaha
Srotodusti	Sanga
Adhisthana	Garbhashaya, Beejagranthi
Rogamarga	Abhyantara
Roga Prakruti	Chirakari

Flowchart No. 1 : Samprapti:[47]



Sapeksha Nidana:

- Artava kshaya explained as a symptom of many yonivyapad. As it is essential to confirm the diagnosis beforehand of treatment, the differentiation of condition with similar pathology is done as below.

1. **Vata yonivyapad:** In this condition, lakshanas like ayama, suptaa, pipilika sarana, yonikarkashata are seen which are not concerned for artavakshaya.[48]
2. **Arajaska yonivyapad:** Symptoms like daha, karshya, vaivarnya are in Arajaska.[49]
3. **Vandhya:** It is nashtartava but not ksheenartava.[50]
4. **Shandi:** Stana abhava is factor for differentiation.[51]
5. **Lohita kshaya**: In this condition, ksheena rajas with daha etc are profoundly observed.[38]

Sadhya-Asadhyata

- In Ayurvedic classics, prognosis of Artava Kshaya is not mentioned but Sushruta explained treatment of Artava Kshaya.
- By which it can be interpreted that Artava Kshaya is either Sadhya or Kashta-Sadhya Vyadhi.

Chikitsa

- Chikitsa is **Samprapti Vighatana**. In Artava Kshaya, there is vitiation or Vruddhi of Vata and Kapha along with Kshaya of Pitta, Rasa and Rakta Dhatu due to Agnimandya. Hence all these Samprapti Ghataka should be considered to treat Artava Kshaya.
- **Nidana Parivarjana** is the basic management of all diseases. In Artava Kshaya, Hetus should be avoided in order to get permanent relief.

- Artava Kshaya should be treated with Samshodhana with Agneya Aushadhi.[52] Dalhana explained Shodhana should be done by Vamana and not by Virechana. Because Virechana reduces Pitta that reduces Artava. Vamana eliminate Saumya Dhatu-Kapha resulting into a relative increase in Agneya Dhatu of body, consequently Artava quantity increases.[53]
- As per Chakrapani Tika - Purifying measures clear Srotasa (Channels). Vamana Karma and Virechana Karma clears upward and downward directed channels respectively, thus both should be used, giving due consideration to the dosages of drugs used for purification and fitness of the patient.[54]
- In Artava Kshaya there is qualitative Kshaya of Pitta and quantitative Kshaya of Artava. The use of Agneya Dravyas is beneficial in Artava Kshaya because of Agnivardhaka and Pittakara properties that helps in Samprapti Vighatana.

Modern View

Anatomical Aspect,

The major component of female reproductive system are vulva, vagina, uterus, fallopian tubes & ovaries.

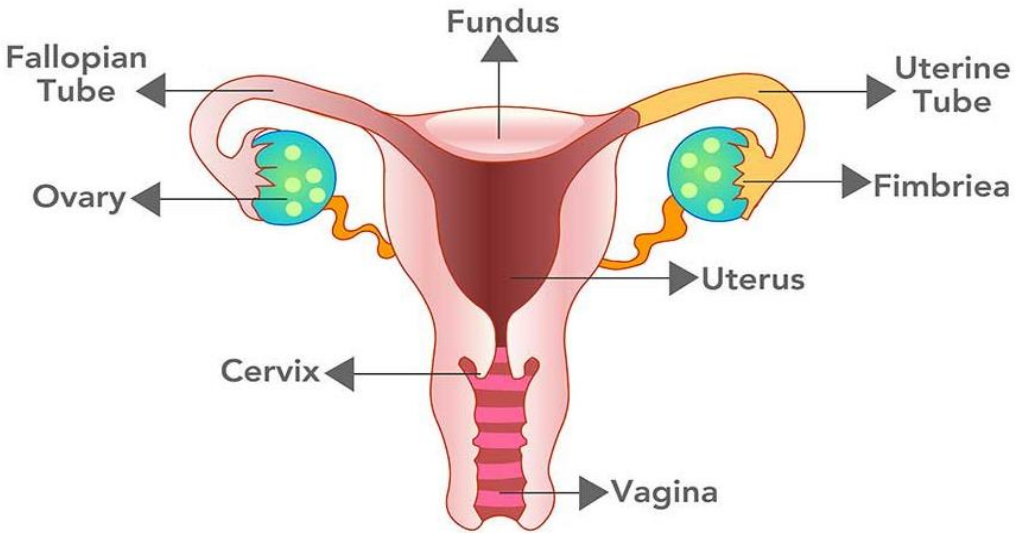


Figure – 1: Anatomical Structure of Uterus

Anatomy of uterus[55]

The uterus is a hollow pyriform muscular organ situated in the pelvis between the bladder in front and the rectum behind.

Position: Its normal position is one of the anteversion and anteflexion.

Measurements: The uterus measures about 8 cm long, 5 cm wide at the fundus. walls are about 1.25 cm thick. Weight varies from 50–80 g.

Parts:

I. **Body or corpus:** The body is further divided into fundus the part which lies above the openings of uterine tubes. The body is triangular and lies between the openings of the tubes and the isthmus. The superolateral angles of the body of the uterus project outwards from the junction of the fundus and body and are called the cornua of the uterus. The uterine tube, round ligament, and ligament of the ovary are attached to each cornu.

II. **Isthmus:** It is a constricted part measuring about 0.5 cm situated between the body and cervix.

III. **Cervix:** It is the lowermost part of the uterus. It extends from the histological internal os and ends at external os which opens into the vagina after perforating the anterior vaginal wall. It is almost cylindrical in shape and measures about 2.5 cm in length and diameter.

It is divided into a supravaginal part—the part lying above the vagina and a vaginal part which lies within the vagina, each measuring 1.25 cm.

In nulliparous, vaginal part of the cervix is conical with external os looking circular, where as in parous, it is cylindrical with external os having bilateral slits.

IV. **Cavity:** The cavity of the uterine body is triangular on coronal section with the base above and the apex below. It measures about 3.5 cm. There is no cavity in the fundus. The normal length of the uterine cavity including the cervical canal is usually 6–7 cm.

Structures:

Body-The wall consists of 3 layers from outside inwards:

I. **Perimetrium-** It is the serous coat which invests the entire organ except on the lateral borders. The peritoneum is intimately adherent to the underlying muscles.

II. **Myometrium-** It consists of thick bundles of smooth muscle fibers held by connective tissue and are arranged in various directions. During pregnancy, however, three distinct layers can be identified-outer longitudinal, middle interlacing and the inner circular.

III. **Endometrium-** It is the mucous lining of the cavity. As there is no sub mucous layer it is directly opposed to the muscle coat. It consists of lamina propria that contains stromal cells, endometrial glands, vessels and nerves and surface epithelium which is a single layer of ciliated columnar epithelium.

The endometrium consists of following three layers from outside to inside these are:

a) Stratum basale / basal layer: It is thin and has a separate blood supply.

b) Stratum spongiosum/spongy layer: It is thick and edematous.

c) Stratum compactum/compact layer: It is thin and superficial towards the uterine lumen. It consists of compactly arranged stromal cells.

Blood Supply:

Arterial supply: The arterial supply is from the uterine artery- one on each side.

The other sources are ovarian and vaginal arteries to which the uterine arteries anastomose.

Veins: The venous channels correspond to the arterial course and drain into internal iliac veins.

Nerve Supply:

The nerve supply of the uterus is derived principally from the sympathetic system and partly from the parasympathetic system.

The somatic distribution of uterine pain is that area of the abdomen supplied by T10 to L1. The parasympathetic system is represented on either side by the pelvic nerve which consists of both motor and sensory fibers from S2, S3, S4.

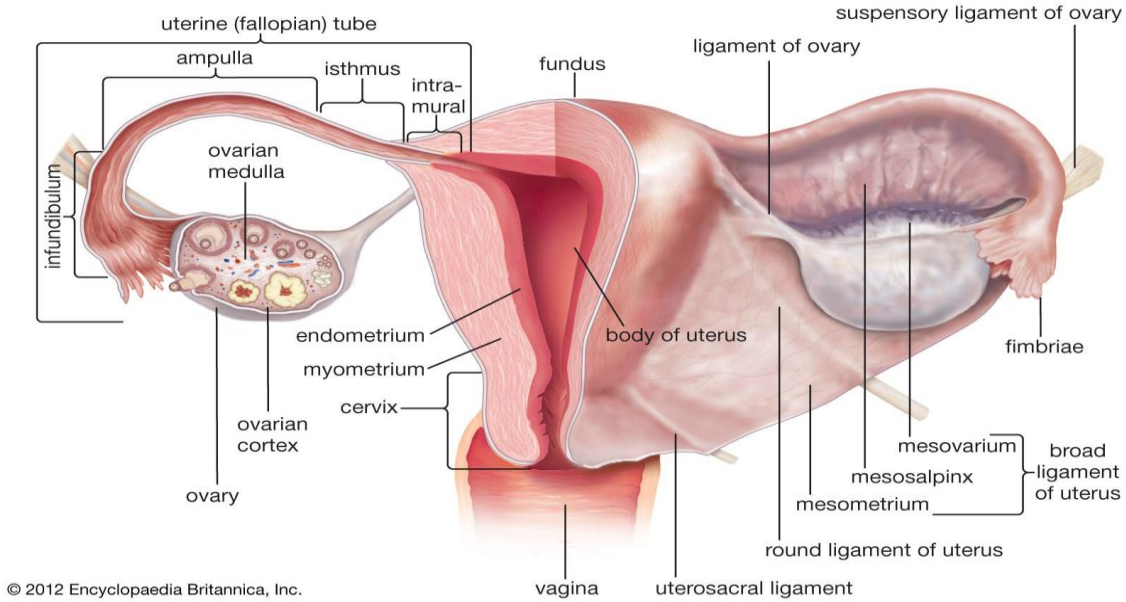


Figure – 2: Anatomical Structure of Female genital tract

MENSTRUATION[56]

Menstruation is the visible manifestation of cyclic physiological uterine bleeding due to shedding of endometrium following invisible interplay of hormones mainly through the hypothalamo – pituitary – ovarian axis (H-P-O axis).

For the menstruation to occur:

• The axis must be actively coordinated.

• Endometrium must be responsive to the ovarian hormones.

• The outflow tract must be patent.

Functions of Ovaries – Ovaries are the primary sex organs in females with functions:

Secretion of female sex hormone

Oogenesis

Menstrual cycle

Ovarian hormones – Ovary secretes the female sex hormones mainly estrogen and progesterone and also produce inhibin relaxin and small quantities of androgens.

Estrogen & progesterone which are secreted by follicles and corpus luteum. Ovarian follicles secret a large quantity of estrogen.

Age of menarche: The first menstruation occurs between 11 to 15 yrs with a mean of 13 years. It is more closely related to bone age than to chronological age, for the last couple of decades, the age of menarche is gradually declining with improvement of nutritional and environmental condition.

Age of menopause: Menstrual cycle ceases at the age of 45 to 50 yrs. Permanent cessation of menstrual cycle in old age is called menopause.

Once the menstruation starts it continues cyclically at intervals of 21-35 days with a mean of 28 days.

Physiologically it is kept in abeyance due to pregnancy and lactation.

The duration of menstruation is about 4-5 days and amount of blood loss is estimated to be 20 to 80 ml with an average of 35ml.

Nearly 70% of total blood loss occurs in first two days.

The menstrual discharge mainly consists of dark altered blood mucus, vaginal epithelial cells, fragments of endometrium, prostaglandins, enzymes and bacteria.

Menstrual cycle:

- Ovarian cycle
- Endometrial cycle

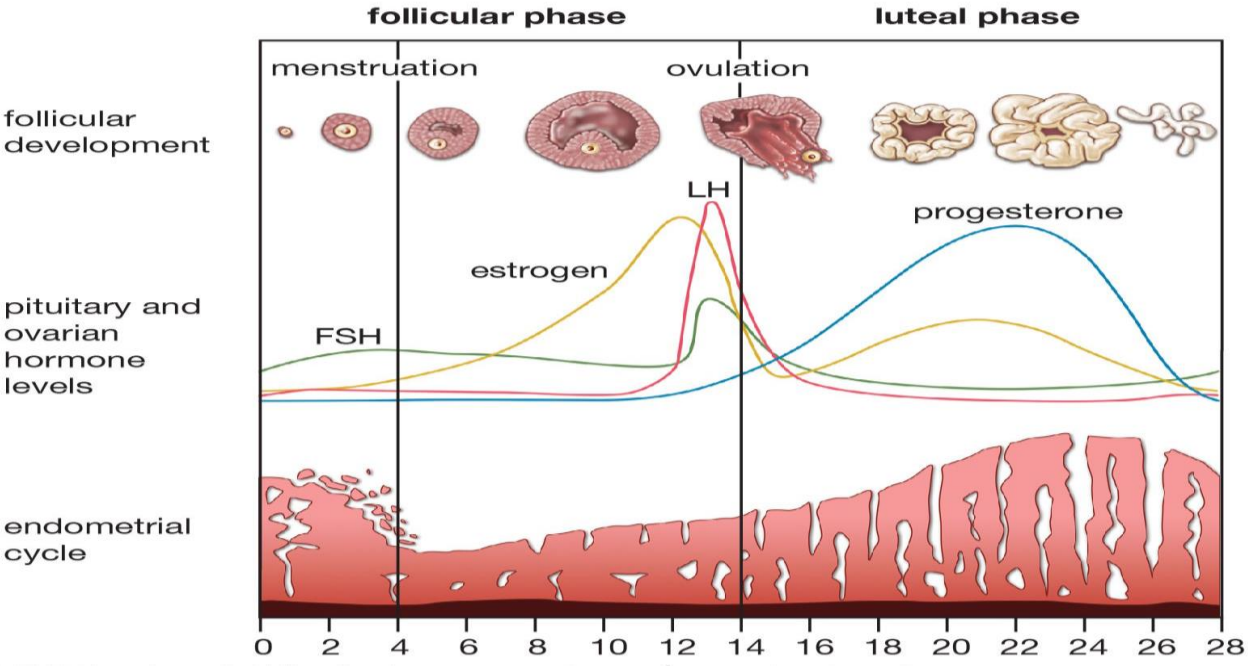


Figure – 3: Phases of Menstrual cycle

Ovarian cycle: The development and maturation of a follicle, ovulation and formation of corpus luteum and its degeneration constitute an ovarian cycle.

The ovarian cycle consists of:

• **Recruitment of groups of follicles:** The cohort of the growing follicles undergoes a process of development and differentiation which takes about 85 days and spreads over 3 ovarian cycles. It is presumed that about 20 antral follicles (about 5–10 per ovary) proceed to develop in each cycle. The initial recruitment and growth of primordial follicles are not under the control of any hormone.

After a certain stage (2–5 mm in size) the growth and differentiation of primordial follicles are under the control of FSH. Unless the follicles are rescued by FSH at this stage, they undergo atresia.

• **Selection of the dominant follicle:** As early as day 5–7, one of the follicles out of so many becomes dominant and undergoes further maturation. It seems probable that the one with highest antral concentration of estrogen and lowest androgen: estrogen ratio (estrogenic microenvironment) and whose granulosa cells contain the maximum receptors for FSH, becomes the dominant follicle. The rest of the follicles become atretic by day 8. The granulosa cells surround the ovum to form cumulus oophorus. The cells adjacent to the ovum are arranged radially and is called corona radiata. The fully mature Graafian follicle just prior to ovulation measures about 20 mm.

• **Ovulation:** The dominant follicle, shortly before ovulation reaches the surface of the ovary. The cumulus becomes detached from the wall, so that the ovum with the surrounding cells (corona radiata) floats freely in the liquor folliculi. The oocyte completes the first meiotic division with extrusion of the first polar body which is pushed to the perivitelline space. The follicular wall near the ovarian surface becomes thinner. The stigma develops as a conical projection which penetrates the outer surface layer of the ovary and persists for a while as a thin membrane. The cumulus escapes out of the follicle by a slow oozing process, along with varying amount of follicular fluid. The stigma is soon closed by a plug of plasma. The ovum is picked up into the fallopian tube and undergoes either degeneration or further maturation, if fertilization occurs.

• **Corpus luteum formation:** After ovulation, the ruptured Graafian follicle develops into corpus luteum. The life cycle is divided into four stages:

a) **Stage of Proliferation**

b) **Stage of Vascularization**

c) **Stage of Maturation**

d) **Stage of Regression**

Endometrial cycle:

The endometrium is the lining epithelium of the uterine cavity above the level of internal os. It consists of surface epithelium, glands, stroma and blood vessels. Two distinct divisions are established.

a) Basal zone: It is about one-third of the total depth of the endometrium and lies in contact with the myometrium. It consists of stromal cells. The base of the endometrial glands extends into the layer. The zone is supplied by the basal arteries. The zone is uninfluenced by hormones. After shedding of the superficial part during menstruation, the regeneration of all the components occurs from this zone. It measures about 1 mm.

b) Functional zone: This zone is under the influence of fluctuating cyclic ovarian hormones, oestrogen and progesterone. The cyclic changes have been divided into four phases.

i. Stage of regeneration: Regeneration of the endometrium starts even before the menstruation ceases and is completed 2–3 days after the end of menstruation. New blood vessels grow from the stumps of the old one. The glands and the stromal cells are regenerated from the remnants left in the basal zone. The average thickness is 2 mm.

ii. Stage of proliferation: This stage extends from 5th or 6th day to 14th day. The proliferative changes occur due to rise in level of ovarian estrogens. There is proliferation of all the elements. The glands become tubular. The epithelium becomes columnar. The stromal cells become spindle shaped. The spiral vessels extend unbranched to a region below the epithelium. The thickness measures about 3–4 mm.

iii. Secretory phase: The changes of the components are due to the combined effects of oestrogen and progesterone. Growth begins on day 15 and ceases 5-6 days prior to menstruation. This is due to dehydration of the glands. The regressive changes in the endometrium are pronounced 24 – 48 hours prior to menstruation. The thickness of the endometrium reaches its highest (6 – 8 mm).

iv. Menstrual phase: It is essentially degeneration and casting off an endometrium prepared for a pregnancy. Regression of corpus luteum with fall in the level of oestrogen and progesterone is an invariable preceding feature. As a result of withdrawal of hormone support, there is retrogressive changes in the endometrium.

Mechanism of menstrual bleeding:

The degenerative changes are predominantly of vascular origin. Stasis of blood and spasm of the arterioles lead to damage of the arteriolar walls. Phase of relaxation leads to escape of blood out of the vessels through the damaged walls. The degenerative process is rapid and involves all the components of the functional damaged layer. There is local tissue destruction by release of proteolytic enzymes from the breakdown of lysosomes. So, there is enzymatic autodigestion of the functional zone. The bleeding occurs from the broken arteries, veins and capillaries and also from the stromal hematoma. The blood along with the superficial functional layer is shed into the uterine cavity. The menstrual flow stops as a result of combined effect of prolonged vasoconstriction, myometrial contraction and local aggregation of platelets with deposition of fibrin around them. Endothelin and platelet activating factor present in the endometrium are potent vasoconstrictors. Resumption of oestrogen secretion leads to clot formation over the decapitated stumps of endometrial vessels. There is simultaneous repair of endometrium. This is under control of oestrogen and different growth factor.

Role of prostaglandins:

The arteriolar constriction and endometrial necrosis are caused by prostaglandins. The endometrium and partly the myometrium, synthesize the prostaglandins from arachidonic acid by the enzyme cyclo-oxygenase. PGF2α causes myometrial contraction and vasoconstriction. It plays a dominant role in normal cycle. PGE2 produces myometrial contraction but causes vasodilatation. PGI2 (Prostacyclin) causes myometrial relaxation and vasodilatation. It also inhibits platelet activity. Thus, the menstrual pain and blood flow are probably related to the relative proportion of different prostaglandins present in the endometrium.

HYPOMENORRHEA

Hypomenorrhea also known as short or scanty periods, is extremely scanty or less menstrual blood flow. Uterine bleeding is slight or less in amount or duration or both. Bleeding that lasts for less than 2 days is termed as Hypomenorrhea.[57]

CAUSES:

Constitutional:

Many times, scanty menses is found in the whole menstrual cycle and is of no clinical significance. It is perhaps due to presence of unusual insensitive arrangement of endometrial vascular apparatus.

Uterine:

It may follow myomectomy, intrauterine adhesions, or any other operations on the uterus where endometrial cavity is reduced. Endometrial tuberculosis forms 50% case of hypomenorrhea in women, hence it is vital to perform preliminary hysteroscopic examination in hypomenorrhic before attempting IVF. A positive case of Tb is managed by lysis of adhesions followed by IUCD insertion.

Hormonal:

Hypomenorrhea may precede amenorrhea. It may also happen in cases with long term use of low dose OCP leading to progress endometrial atrophy.

Nervous and emotional:

Psychological factors failing to supress ovarian and uterine cycle completely succeed only in reducing the amount of flow as seen in pseudocyesis where hypomenorrhea is more common than amenorrhea.

PATHOPHYSIOLOGY:

Pituitary dysfunction → Unresponsive FSH → Ovarian unresponsiveness → hypomenorrhea.

Undue prolongation of proliferation phase with normal secretory phase.

Criteria For Diagnosis:

- Menstrual bleeding occurring more than 35 days apart
- Remains at the same frequency
- Unduly scanty
- Menstrual flow lasting for less than 2 days.

Examination:

General examination, abdominal examination and pelvic examination is mandatory in all women having hypomenorrhea.

Investigations:

- Urine pregnancy test to exclude pregnancy
- General routine check-up blood, urine and stool to rule out any infection.
- Per speculum and per vaginal examination.
- Hormonal assessment for FSH, LH, serum PRL, testosterone, thyroid levels
- Ultrasonography
- C.T scan
- Laparoscopy
- Chromosomal study

TREATMENT:

Mainly depends on the cause

- Reassurance and psychotherapy are very useful in psychogenic and environmental cause.
- Anaemia should be treated with appropriate medications
- In case of malnutrition – management with proper diet
- In case of obesity – weight loss therapy, treating the cause
- Systemic illness should be managed adequately
- Hormonal therapy & surgical therapy.

DRUG REVIEW

DRUG REVIEW

According to brihatrayi, the one of the significant factors in treatment is aushadha later to bhishak. Drug has its specific role in treatment aspect of diseases, it is included one among chikitsa chatushpada (Bhishaka, Dravya, Upasthata, Rogi). In Ayurvedic classics, action of each drug is described based on its rasa, guna, virya, vipaka, prabhava.

Among all the preparations, “Lashuna Kalpa” And “Rasnadi Tailam” is mentioned in Ashtanga sangraha uttarasthana which was selected as trial drug for carrying over present study on Artavakshaya.

❖ LASHUNA KALPA

Ingredient: -

- Rasona (Lashuna Clove)

❖ RASNADI TAILAM

रास्नामालती च्छिन्नरुहा मधुकबलाव्याघ्रीदेवदारुचित्रकयूथिकामूलैः कर्षांशैस्तैलप्रस्थः समूत्रो द्विगुण पयसा विपक्वः पिचुना प्रणीतोऽनिलार्तिहरः।

(अ.सं.उ. 39/36)[58]

Ingredients:

I. Kalka dravya:

Table no - 6: Kalka Dravya of Rasnadi taila

Rasna	Yashtimadhu	Devadaru
Malati	Bala	Chitraka
Guduchi	Kantakari	Yuthika

II. Drava dravya: Jala, Gomutra & Godugdha (2 times more in quantity than oil)

III. Sneha dravya: Tila Tailam (Moorchita)

Table no - 7: LASHUNA[59]

Latin name	Allium Sativa  **Figure 4 : Lashuna**
Family	Lilliaceae
Synonyms	Ugragandha, Yavaneshtha, Rasonaka, Malechchhukanda
Part used	Clove & Oil
Vernacular names	**English -** Garlic **Hindi –** Lashuna
Classification	Charaka – Harita Varga
Rasapanchaka	**Rasa**- Amla rahita Pancha rasa **Guna** - Snigdha, Sara, Teekshna, Guru, Pichchhila **Virya**- Ushna **Vipaka** - Katu
Effect on Dosha	Vata kapha-hara
Indications	Yonivyapada, Sotha, Vatavyadhi, Rajyakshma, Gulma, Vibandha, Pakshaghata, Agnimandya, Aruchi, Ajeerna, Vibandha, Shoola, Krimi, Arsha, Hridroga, Jeernakasa, Shwasa, Swarabheda, Shukradaurbalya, Kashtartava, Asthibhagna, Jeernajwara, Vishamajwara
Actions	Artava janana, Sothahara, Vedanasthapana, Vishaghna, Raktoktkleshaka, Rasayana, Shulaprashaman, Raktapitta janana, Krumighna, Dipana, Pachana, Anulomana, Medhya, Netrya, Krimighna, Hridayottejaka, Kaphanissaraka, Kanthya, Mutrajanana, Shukra anana, Amapachana
Chemical constituents	Allin, Allicin Mucilage, Albumin, Five saponins, Diallyl trisulphide, Allylpropyl disulphide, Glucofructan, S-amino acids
Pharmacological activities	**Uterine Stimulant**, Anti-bacterial, Anti-Fungal, Fibrinolytic, Antitumor, Anticoagulant, Anti-inflammatory, Hepatoprotective, Larvicidal, Free-Radical scavenging, Hypolipidemic, Hypoproteinemic, Hypocholesteremic, Antihypertensive, Anticancer, Insecticidal, Antitubercular, Antioxidant

Table no – 8: RASNA[60]

Latin name	Pluchea Lanceolata  **Figure 5 : Rasna**
Family	Asteraceae
Synonyms	Elaparni, Surabhi, Suvaha, Gandhmoola, Gandhanakuli, Yuktarasa, Yukta, Surasa, Shreyasi, Nakuli
Part used	Patra, Moola
Vernacular names	**English –** Lesser Galangal **Hindi -** Rayasan
Classification	Anuvasanopaga, Vayastapana, Arkadi gana, Sleshmashamshamana, Oshadi varga, Guduchyadi varga, Haritakyadi varga, Pippalyadi varga
Rasapanchaka	**Rasa** - Tikta **Guna** - Guru **Virya** - Ushna **Vipaka** - katu
Effect on Dosha	Kapha - Vata Shamaka
Indications	Vatavyadhi, Gridhrasi, Amadosha, Gulma, Udarashoola, Vibandha, Vatarakta, Raktavikara, Kasa, Shwasa, Shotha, Shoola, Vishavikara, Arsha
Actions	Shothahara, Sheetaprashamana, Vedanasthapana, Raktashodhaka, Shoolaprashamana, Vrishya, Rasayana, Rechana
Chemical constituents	Moretenol, neolupenol, hexacosanoictetracosa noic acid, betasistosterol, D glucoside, Pluchine, Quercetin
Pharmacological activities	Anti-inflammatory, Antibiotic, Immunostimulant, Antioxidant

Table no – 9: MALATI[61]

Latin name	Jasminum officinale  **Figure 6 : Malati**
Family	Oleaceae
Synonyms	Hrudya Gandha, Manojna, Ratripushpa, Varshabhava, Rajaputri, Surapriya
Part used	Moola, Patra, Pushpa
Vernacular names	**English –** Spanish Jasmine **Hindi -** Chameli
Classification	Kushthaghna, Pushpa varga, Amradi varga, Oushadhi Varga, Karaveeradi Varga
Rasapanchaka	**Rasa**- Tikta, Kashaya **Guna** - Laghu,Snigdha, Mrudu **Virya**- Ushna **Vipaka** - Katu
Effect on Dosha	Tridosa-shamaka
Indications	Tridoshajavikara, Dantadaurbalya, Mukharoga, Shirahshoola, Shoola, Pakshaghata, Ardita, Vatavikara, Manasadaurbalya, Dhwajabhanga, Kandu, Kushtha, Twagdosha, Mootraghata, Rajorodha, Vrana, Udavarta, Raktavikara, Mootrakrichchhra, Vishavikara
Actions	Artavajanana, Saumanasyajanana, Medhya, Vajikarana, Vranaropana, Vranashodhaka
Chemical constituents	Benzylacetate, linalool, linalylacetate, benzylalcohol, indole, jasmone, methyl anthranilate, P-cresol, geraniol, racemic (5-pent- 2 enyl)-5, 1-pentanolide, benzyl benzoate, nerol, l-a-terpineol, d- and dl- linalool, y- jasmolactone
Pharmacological activities	Anticancer, Hypotensive, Diuretic, Antibacterial, Anti-inflammatory

Table no – 10: GUDUCHI[62]

Latin name	Tinospora cardifolia  **Figure 7 : Guduchi**
Family	Menispermiaceae
Synonyms	Amrita, Tantrika, Kundalini, Vatsodini, Chhina, Amritavalli, Madhuparni
Part used	Panchanga
Vernacular names	**English –** Gulancha Tinospora **Hindi –** Giloy, Guloh
Classification	Vayasthapana, Daha-prashshamana, Trishna nigrahana, Triptighna, Stanya-shodhaka, Guduchyadi,, Patoladi gana, Valli panchmoola, Kakolyadi, Aragvadhadi
Rasapanchaka	**Rasa**- Tikta, Kashaya **Guna** - Laghu, Ruksha **Virya**- Ushna **Vipaka** - Madhura
Effect on Dosha	Tridosa-shamaka
Indications	Kushtha, Vatarakta, Trishna, Daha, Chhardi, Aruchi, Agnimandya, Shoola, Yakridvikara, Kamala, Amlapitta, Pravahika, Atisara, Raktatisara, Grahani, Krimi, Arsha, Hriddaurbalya, Pleehavriddhi, Vastishotha, Pandu, Shukradaurbalya, Prameha, Madhumeha, Mootrakrichchhra, Twagroga, Phiranga, Jwara
Actions	Vedanasthapana, Deepana, Pachana, Rasayana, Raktashodhaka
Chemical constituents	Terpenoids, Alkaloids Tinosporine, Berberine, Lignans 3(a-4-dihroxy-3- methoxybenzyl), Steroids, (Gilolinsterol, β-sitosterol, 20-α Hydroxy ecdysone), Magnoflorine, palmetine
Pharmacological activities	Antioxidant, Antidiabetic, Hypolipidemic, Anticancer, Immunomodulatory

Table no – 11: YASHTIMADHU[63]

Latin name	Glycyrrhiza glabra  **Figure 8 : Yashtimadhu**
Family	Fabaceae
Synonyms	Madhuka, Klitaka, Yashtyahva, Madhuyashti, Jeshtamadhu, Mulahatti
Part used	Moola
Vernacular names	**English -** Liquorice **Hindi -** Mulethi
Classification	Jeevaneeya, Sandhaneeya, Varnya, Kanthya, Kandughna, Chhardinigraha, Shonitasthapana, Mutravirajaneeya, Snehopaga, Vamanopaga, Asthapanopaga, Kakolyadi, Anjanadi, Sarivadi, Haritakyadi
Rasapanchaka	**Rasa**- Madhura **Guna** - Guru, Snigdha **Virya**- Sheeta **Vipaka** - Madhura
Effect on Dosha	Vata - pitta shamaka
Indications	Vranashotha, Visha, Khalitya, Palitya, Udarashoola, Raktapitta, Shwasa, Kasa, Swarabheda, Urogata vrana, Mootrakrichchhra, Pooyameha, Paittika prameha, Shukrameha,
Actions	Dahashamak, Keshya, Vedanastapana, Shonitastapana, Shukravardhaka, Jeevaneeya, Rasayana, Shothahara, Dahashamaka
Chemical constituents	Glycyrrhizine, Prenylated biaurone, licoagrone, 7-acetoxy-2-methyl- isoflavone, Liqcoumarin, Isoflavone, Glyzaglabrin, Astragalin, Liquiritigenin and Isoliquiritigenin, Glycerol, Licoricone
Pharmacological activities	Smooth muscle depressant, Anti-microbial, Hypolipidaemic, Antiantherosclerotic, Antiviral, Hypotensive, Hepatoprotective, Anti-exudative, Spasmolytic, Antidiuretic, Antiulcer, Antipyretic, Antioxidant, Anti-inflammatory

Table no – 12: BALA[64]

Latin name	Sida cordifolia  **Figure 9 : Bala**
Family	Malvaceae
Synonyms	Vatyapushpi, Badraudani, Kharayashtika, Maha, Samanga
Part used	Moola
Vernacular names	**English –** Country mallow **Hindi –** Bariyaar, Kungyi
Classification	Madhuraskandha, Bruhaniya, Prajasthapana, Vatasamana
Rasapanchaka	**Rasa**- Madhura **Guna** - Guru, Snigdha, Pichchila **Virya** - Sheeta **Vipaka** – Madhura
Effect on Dosha	Tridosa-shamaka
Indications	Vatavyadhi, Pakshaghat, Raktapitta, Rajayakshma, Pradara, Garbhashaya Daurbalya, Yoniroga, Sutikaroga, Mootrakrichchhra, Jwara, Galaganda
Actions	Balya, Prajasthapana, Ojovardhaka, Brihana, Vatasanshamana, Shukrala, Mootrala, Jwaraghna
Chemical constituents	Ephedrine, Hypaphorine, vascinone
Pharmacological activities	Anti- inflammatory, Analgesic, Antioxidant, Antispasmodic, Thyroregulatory

Table no – 13: KANTAKARI[65]

Latin name	Solanum surrattense  **Figure 10 : Kantakari**
Family	Solanaceae
Synonyms	Duhasparsha, Kshudra, Vyaghri, Rashtrika, Kuli, Dhavani, Kshudraphala, Chitraphala, Nidigdhika, Dravani
Part used	Panchanga
Vernacular names	**English –** Yellow – berried nightshade **Hindi –** Kateli, Katai
Classification	Kasahara, Shothahara, Hikkanigrahana, Kanthya, Angamardaprashamana, Brihatyadi, Varunadi, Guduchyadi
Rasapanchaka	**Rasa**- Katu, Tikta **Guna** - Laghu, Ruksha **Virya** - Ushna **Vipaka** - Katu
Effect on Dosha	Kapha - Vata Shamaka
Indications	Pinasa, Angamarda, Sandhivata, Agnimandya, Udavarta, Krimi, Raktavikara, Mootrakrichchhra, **Rajorodha, Kashtaprasava, Klaibya**, Charmaroga
Actions	Vedanastapana, Shothahara, Raktashodhaka, Kasahara, Kanthya, Hikkanigrahana, **Garbhashaya sankochaka**, Vajikarana
Chemical constituents	Carpesterol, glucoalkaloid, steroidal compounds, solasonine, solamargine, apigenin, solanocarpine, esculin, solasonine, Glucoside, alkaloids, flavonoids, ascorbic acid
Pharmacological activities	Antibiotic, Vasoconstrictor, Hypocholesterolaemic, Hepatoprotective, Antiandrogenic, Nephroprotective, Anti-inflammatory, Analgesic, Stimulant, Appetizer, Antioxidant, Diuretic, Anti Urolithiatic

Table no – 14: DEVADARU[66]

Latin name	Cedrus deodara  **Figure 11 : Devadaru**
Family	Pinaceae
Synonyms	Indradaru, Bhurah, Bhadrakastha, Suradaru, Drunklina
Part used	Bark
Vernacular names	**English –** Deodar, Himalayan cedar **Hindi –** Debdar, Deyodar
Classification	Stanya shodhana, Anuvasanopaga, Vata samshamana
Rasapanchaka	**Rasa** - Tikta **Guna** - Laghu, Snigdha **Virya** - Ushna **Vipaka** - Katu
Effect on Dosha	Kapha - vata shamaka
Indications	Sandhivata, Shotha, Vedana, Twagvikara, Kandu, Vrana, Kshata, Vatarakta, Gridhrasi, Karnashoola, Shirahshoola, Amadosha, Adhamana, Vibandha, Vatavikara, Krimiroga, Hrid-daurbalya, Kaphaja gandamala, Galaganda, Shleepada, Upadamsha, Jeernakasa, Peenasa, Hikka, Mootrakrichchhra, Pooyameha, Stanyadosha, Sutikaroga, Medoroga, Jalodara
Actions	Sothaghna, Lekhana, Garbhashaya Shodhana, Stanyashodhana
Chemical constituents	Terpinoids Flavonoids Lignanemixture P-Methyl acetophenone, 3,4- dimetho-xiphenyl, Tannins, Haemochalol, Cedrinoside, Cedrusinine, Cedrol
Pharmacological activities	Anti-inflammatory, Spasmolytic, Immuno modulatory, Neuroleptic Activity, Anticonvulsant, Anti diabetic

Table no – 15: CHITRAKA[67]

Latin name	Plumbago zeylanica  **Figure 12 : Chitraka**
Family	Plumbaginacae
Synonyms	Anala, Agni, Chitrapali, Katushiki, Pavaka, Dweepi Vyala, Pithi
Part used	Mula twak
Vernacular names	**English –** Cyelon leadwort **Hindi –** Chita, Chitra
Classification	Deepaniya, Triptighna, Shoola prashamana, Bhedaniya, Lekhaniya, Pippalyadi, Varunadi, Panchakola, Shadushana
Rasapanchaka	**Rasa**- Katu **Guna** - Laghu, Ruksha, Teekshna **Virya**- Ushna **Vipaka** - Katu
Effect on Dosha	Kapha - vata hara, Pitta vardhana
Indications	Shotha, Switra, Shleepada, Amavata, Nadidaurbalya, Vatavyadhi, Agnimandya, Ajirna, Udarashoola, Yakridvikara, Arsha, **Rajorodha**, Prasutivikara
Actions	Grahi, Krimighna, Garbhashaya sankochaka, Garbhasravakara, Sweda janana
Chemical constituents	Plumbagin, Isozcylanon, zeylanon, elliptinone, droserone, 5coumarins-seselin, xanthoxyletin, 3-chloroplumbagin, 3- biplumbagin, binaphthoquinone
Pharmacological activities	**Uterotonic**, Anticarcinogenic, Antitumor, Anti-diabetic, Antiplasmodial, Antimicrobial, Antibiotic, Antifungal, Anti-inflammatory, Abortifacient, Anti-fertility, Antioxidant, Hepatoprotective, Neuroprotective, Memory inducing, Wound healing

Table no – 16: YUTHIKA[68]

Latin name	Jasmin Auriculata  **Figure 13 : Yuthika**
Family	Oleaceae
Synonyms	Sumana, Vasanthi Punyagandha, Jati, Bala, Balapushpa, Kandi, Charumoda, Shikhandini, Pushpagandha
Part used	Moola
Vernacular names	**English -** Jasmine **Hindi -** Juhi
Classification	Amradi Varga, Pushpa Varga, Oushadhi Varga, Karaveeradi Varga
Rasapanchaka	**Rasa**- Tikta, Kashaya **Guna** - Laghu **Virya**- Sheeta **Vipaka** - Katu
Effect on Dosha	Pitta shamak
Indications	Vrana, Mukharoga, Dantaroga, Akshiroga, Visha, Daha, Trushna
Actions	Raktashodhaka, Vranaropana, Vishaghna, Arshaghna, Truptikara
Chemical constituents	Lupeol, Hentriacontane, n- tricantanol, jasminol, d- mannitol, jasmine
Pharmacological activities	Anti lithiatic, Anti-inflammatory, Anti-ulcer

Table no – 17: TILA TAILA[69]: Ayurveda has mentioned Tila Taila as one of the best amongst all herbal oils. Sharangadhara quoted that whenever oil is not specific, sesame oil is to be used.[70]

Latin name	Sesamum indicum  **Figure 14 : Tila**
Family	Pedaliaceae
Synonyms	Tila, Til, Gumbulu, Tara, Ellu.
Part used	Seed-oil
Vernacular names	**English -** Sesame **Hindi -** Til
Rasapanchaka	**Rasa**- Madhura, tikta **Anurasa:** Kashaya **Guna** - Ushna, vyavayi, teekshna, vishada, guru, sara, vikasi, lekhana **Virya**- Ushna **Vipaka** - Madhura
Effect on Dosha	Vata-kaphahara
Indications	Vatashoola, Amavata, Suryavarta, Indralupta, Mastishkadaurbalya, Atisara, Raktatisara, Agnimandya, Grahani, Arsha, Raktarsha, Raktasrava, Pravahika, Vatarakta. Hikka, Shwasa, Mootrarodha, Rajorodha, Kashtartava, Stanyalpata, Vrana
Actions	Balya, Deepana, Shroto-vishodhana, Tvachya, Vrushya, Vaya-sthapana, Medhya-kara
Chemical constituents	Sesamol, Sesaminol, Vitamin-E, acts as natural antioxidant, Vitamin K, B6, iron, calcium. It has a balanced ratio of omega-3, omega-6, and omega-9 fatty acids
Pharmacological activities	Antioxidant, Antihypertensive, Immunoregulatory and Antitumor, Free Radical Scavenging property

Table no – 18: GODUGDHA:[71]



Figure 15 : Godugdha

RASAPANCHAKA	CHEMICAL COMPOSITION	PHARMACOLOGICAL ACTION
Rasa-Madhura **Guna**-Mrudu, Shlakshana, Balya, Alpaabhishyanda **Virya**-Sheeta **Vipaka**-Madhura **Karma**-Vata-Pittahara, Jeevaniya, Rasayana, Ojuvardhaka.	Diacylglycerol, cholesterol, phospholipids, oligosaccharides, Oleic acid, Myristic acid, N - acetylneuraminic, Immunoglobulins, IgG, IgM, IgA, alpha lactoalbumin, beta lactoglobulin, betacellulin, Insulin like growth factor -I & II.	Antioxidant, Anti-inflammatory, Anti-tumour, Hypocholesterolaemia, Immune boosting.

Table no – 19: GOMUTRA:[72]



Figure 16 : Gomutra

RASAPANCHAKA	CHEMICAL COMPOSITION	PHARMACOLOGICAL ACTION
Rasa - Kashaya, Lavana, Katu, Tikta **Guna** - Laghu, teekshna, ushna **Vipaka** - katu **Veerya -** Ushna **Doshaghnata -** vata-kapha shamaka, pitta - vardhak Balances Vata and Kapha and increases Pitta.	Urea, Phosphrous, Manganese, Carbolic acid, Ammonium nitrogen, Chloride, Creatinine, Potassium, Uric acid, Allantoin, Aurum hydroxide, Salt, Sodium, Sulfur, Vitamins A, B, C, D, E	Anti-bacterial, Anti-Fungal, Anthelmintic, Anti-cancer, Immunomodulator

MATERIALS & METHODS

MATERIALS AND METHODS

MATERIALS

- **Lashuna kalpa**

  Lashuna clove dried in shade and made into fine powder (Churna) with the help of grinder & administered orally for 7 days after menstruation.

  Dosage – 12 gms Twice a day

  Anupana - Kshir or Ghrita or Navaneet



Figure 17 : Lashuna kalpa(Churna)

- **Rasnadi Taila**

रास्नामालती च्छिन्नरुहा मधुकबलाव्याघ्रीदेवदारुचित्रकयूथिकामूलैः कर्षांशैस्तैलप्रस्थः समूत्रो

द्विगुण पयसा विपक्वः पिचुना प्रणीतोऽनिलार्तिहरः।

(अ.सं.उ. 39/36)[58]

Ingredients –

Kwatha & Kalka dravyas- Rasna, Yashtimadhu, Devadaru, Malati, Bala, Chitraka, Guduchi, Kantakari, Yuthika

Drava dravya- Jala, Gomutra & Godugdha (2 times more in quantity than oil)

Sneha dravya- Tila Tailam (Moorchita)

Method of Preparation:[73]

Taila murchana: Murchhana alters the solubility pattern and absorbability, which is desired to get maximum medicinal properties; for this oil is boiled till it attain fumes confirming it is devoid of ama.[74]

The kwatha was prepared with above all kalka dravyas according to sharangdhara explanation and mixed with murchhita tila taila in a stainless-steel vessel, on low flame, then kalka was prepared with above all drugs and was added in it.

Then Godugdha and Gomutra was added and stirred. Heated it over Mandagni with continuous stirring this process was carried for 1 day to get water content evaporate in the preparation.

Sneha siddhi lakshanas of madhyama paka were obtained as per classics.[75]



Figure 18 : Rasnadi Taila Preparation



Figure 19 : Rasnadi Taila

METHODOLOGY

A research study should always be well organised and planned so that a proper blueprint is taken up well in advance to avoid any inaccuracy in the study.

The present study "A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA" is carried out with following administrations.

1. Preparation of drug i.e. Lashuna kalpa & Rasnadi Taila
2. Preparation of a format for case taking
3. Study of concept of Artava Kshaya
4. Collection of data (from classics to update) – literacy
5. Analysis of data (observational study -clinical)
6. Discussion of observed findings
7. Interpretation of findings
8. Conclusion of the study

Ethical clearance:

The topic of the present study hypothetically was presented to institutional ethical committee of Dr. N.T.R University of Health Sciences. The significance of study, aim & objectives, methodology & the result anticipated was submitted and clarified to the committee. Ethical clearance was obtained for conduction of the study.

Informed consent:

All the 30 subjects for the study were clearly explained about the procedure they are undertaken, possible responses of the body, safety and confidentiality is clearly assured. An Informed and written consent was obtained.

Objectives of the study:

To evaluate the efficacy of Lashuna kalpa orally and intra-uterine administration of Rasnadi taila in artavakshaya.

To establish the management of hypomenorrhea through Ayurvedic classical procedure called “Uttarabasti”.

Hypothesis

H0 - There is no effect of Lashuna kalpa and Rasnadi taila Uttarabasti in artavakshaya is assumed null hypothesis

H1 – Lashuna kalpa and Rasnadi taila uttarabasti in artavakshaya is effective by regulation of H-P-O axis & improve endometrial response.

MATERIALS

1. Sources of data

a. Literary

b. Sample source

c. Drug source.

a) Literary source:

It is obtained from classical ayurvedic texts, Samhitas, renowned and updated contemporary books of gynaecology, peer reviewed journals and articles related to the present hypothesis and google scholars, PubMed, other web journals and articles.

b) Sample source: A sample size of 30 subjects were selected with diagnosis of artavakshaya from outpatient department of Prasuti Tantra and Stree Roga in Achanta Lakshmipathi Ayurvedic Hospital, Vijayawada. Other available sources were selected for the present study.

c) Drug source: The drug “Lashuna kalpa” and “Rasnadi Taila” was prepared from raw drugs identified and selected from local market and some from coastal region of Vijayawada and the same is prepared in department of rasa shastra and bhaisajya Kalpana, Dr.N.R.S. Government ayurveda College, Vijayawada.

2. Method of collection of data:

An individual case proforma was prepared with in depth history taking, physical signs and symptoms, detailed Dosha, Dushya Pareeksha, necessary laboratory and radiological investigations were procured. The parameters obtained were categorised and scored accordingly. The parameters of signs, symptoms and investigations were scored based on standard method and analysed statistically.

3. Design of study: Randomised clinical study to prove the efficacy of Lashuna kapla and Uttaravasti with Rasnadi taila in artavakshaya in a sample size of 30 subjects.

Diagnostic criteria:

Diagnosis was based upon vyadhi pratyatma lakshana i.e particular cardinal symptoms of artavakshaya w.s.r. to Hypomenorrhea mentioned

- Yathochita kala adarshanam - Intermenstrual period >35 days.
- Alpartava - Duration of bleeding
  decreased amount of menstrual blood.
- Yoni Vedana - Pain in lower abdominal region.

Inclusion criteria: -

- Woman of age group between 18-40 years complaining of scanty menstruation i.e., bleeding for 2-3 days for last 2-3 months will be selected.
- PCOS

Exclusion criteria: -

- Anaemia
- SYSTEMIC ILLNESS- Hypertension, Diabetes Mellitus, Cardiac disease, Genital Tuberculosis, Chronic nephritis, HIV, VDRL, HbsAg positive patients.
- Endocrine disorders – Hyperprolactinemia, Hyperthyroidism
- Androgen producing tumors – Ovarian and Adrenal

Investigations:

- C.B.P.
- R. B.S.
- Thyroid Profile
- Hormonal Assay
- U. S. G – Abdomen & Pelvis
- Viral markers

Intervention:

- 30 patients with cardinal symptoms of artava kshaya are advised to take oral administration of 12 grams of lashuna kalpa BID (in divided doses) with navaneet or ghrita or ksheera and Rasnadi taila Uttar basti followed by purvakarma – shodana basti.
- Uttara basti intra-uterine administration of rasnadi taila 2-5 ml from D6-D10 of menstrual cycle is done.
- The procedure is carried out for 3 consecutive cycles.
- Yonipichu soaked in rasnadi taila is placed in situ for yoni purana.
- Care and concern for intra-procedural hygiene taken.
- Patient was explained for outcome and educated with local, sexual hygiene maintenance.
- Good and nutritious food intake advised post-procedure.
- Regular follow up for improvement of condition is done for every cycle and follow up continued for 2 months after treatment.
- Confidentiality of clinical information of every patient was maintained.

Assessment scale: GRADING

Table no – 20 : Subjective Parameter Scoring

Parameter	Grade 0	Grade 1	Grade 2	Grade 3
Quantity of menstrual flow	2-3 Pads / Day (Normal)	1 Pad / day	Half Pad / Day	Spotting
Duration of menstrual cycle	4-7 Days	3 Days	2 Days	1 Day
Pain in lower abdomen	No Pain	Mild Pain	Moderate Pain	Severe Pain

Objective Parameter Scoring

Table no – 21 : Endometrial Thickness Score

Grade	Incidence	Assessment
0	10-12 mm	Normal
1	13-15 mm	Mild thickened endometrium
2	16-18 mm	Moderate thick endometrium
3	>18 mm	Hyperplasia

Table no – 22 : Scoring on FSH & LH

Grade	Levels of FSH & LH
0	Normal
1	Mild altered
2	Moderately altered
3	Excess altered

Table no – 23 : Overall Results grading

Mild improvement	< 50 %
Moderate improvement	50% to 74%
Good improvement	75% to 100%

UTTARABASTI

उत्तरमार्गदीयमानतया, किंवा श्रेष्ठगुणतया उत्तरवस्तिः । (च.सं.सि. 9/50, चक्र.टी.)[76]

- Cakrapaṇi explains that since it is given in uttara marga (upper passage) i.e. urinary or vaginal introitus, it is called so.
- The Basti given as a paschat karma to the Shodana basti it is termed as Uttarabasti.
- One more term being significant acc to Chakrapani Datta commentary on Charaka siddhi sthana is "Kimva Shrestha Guna Taya uttarabasti". The efficacy of administration being best among all the procedures is named after its effect as "uttarabasti."
- The Uttarabasti has been explained in 2 points of view i.e.
    1. Mutrashaya gata basti – through urinary passage
    2. Garbhashaya gata basti – vaginal route
- Here in, the garbhashaya gata basti in regulating H-P-O- axis is deal with.

Garbhashaya gata basti:[77]

- In this type of Uttarabasti, the basti dravya is administration into uterus via vaginal orifice. This helps in expelling out of Sthanika dosha (vitiated), also corrects the whole uro-genital system through the carried out active principle into the H-P-O axis through the mucosal absorption of the drug.

Uttarabasti – classification:

- The procedure is classified based on many parameters.

i. Based on drug administered

a. Snaihika uttarabasti – Sneha dravya used as medicine for application and there is no additional avapa required for it.

b. Niruha uttarabasti – Kashaya dravya along with Sneha nirvapa is used for administration of uttarabasti.

ii. Based on route of administration

a. Mutrashaya gata – urethral route

b. Yonigata – through vaginal route

c. Garbhashaya gata – intra uterine via cervical canal

Indications of uttarabasti:

- Uttarabasti, being a Sthanika chikitsa has got its limitations for administration, the indication for administering uttarabasti are enumerated as follows
  - Sukradushti
  - Shonita or artava dushti (menstrual abnormalities of female)
  - As a chikitsa for Yonivyapad – like tubal blockages
  - Vandhya chikitsa (to regulate hormonal axis)
  - In retained placenta (apara samsthithi)

Contra- indications:

- Garbhashaya gata Uttara basti in virgin women.
- In rajah kala of a woman

Time of administration:

- The procedure uttarabasti is said to be administered in rutu kala according to classic texts. To be administered only after 2 or 3 asthapana basti as the shodana allows more absorption of the later administered medicine and in rtukala, the dravya is readily received by the yoni marga due to its permeable absorption activity during that period.

Probable mode of action:[78](When administered intravaginally)

- The uttarabasti dravya could act both systemic and local when administered intravaginally. This is through the network of blood vessels existing in the pelvic region say branches of int. Iliac artery, uterine artery and middle rectal and internal pudendal arteries abundantly supply the vaginal walls.
- Therefore, the vascular effect of vaginal tissues causes the uterine pass effect i.e., transfer of vaginal administered drug into the uterus. It is clear through contemporary evidence-based sciences that the higher concentration of progesterone in the uterus is responsible for better absorption of drugs administered intravaginal.

- Thus, it is to be understood that the vaginal canal because of its histopathological support, transports the drug across the vaginal membrane.

    - Transcellular (across epithelial cells) – through diffusion
    - Para cellular (between epithelial cells) – mediated via junctions
    - Vesicular transport (through receptors)

- Therefore, drug absorption in which it is administered through could be absorbed in following manner
    - Dissolution in vaginal layers & membrane penetration
    - Dissolution into vaginal fluids
- Factors that affect the above mechanism could be
    - Physiochemical drug properties
    - Low molecular weight lipophilic form or hydrophilic form
    - Contact time – prolonged contact time increases absorption.

PROCEDURE OF UTTARA BASTI

INSTRUMENTS:

- Sim's posterior vaginal wall speculum
- Anterior wall retractor, vulsellum, uterine sound
- Sponge holding forceps
- Disposable syringe 5ml
- IUI cannula



Figure 20 : Uttar basti tray

PURVA KARMA:

- Patients were given with one course of shodhana basti as purvakarma.
- In the consecutive cycle patients were called on 6th day of menstruation, with a light diet in the morning hours.
- Before giving Uttara basti, blood pressure, temperature and pulse were carefully recorded and noted.
- Abhyanga was done for 30 min in lower abdomen, loin, groin, thigh, buttocks and back, followed by hot water fomentation for 10 mins.

PRADHANA KARMA:

- The patient is placed in lithotomy position.
- Genital parts were cleaned. Per vaginal examination was done to find out the size, shape, and position of uterus.
- Sim's speculum with anterior vaginal wall retractor was applied, cervix visualised and properly held with vulsellum, and uterine sound passed.



Figure 21 : Garbhashayagata Uttarbasti

- The direction & position of the uterus understood by insertion of uterine sound, it also helps to opens the os of cervix, if it is closed.
- Then the 3 ml – 5 ml of medicated taila is taken into the syringe with IUI cannula. Air is removed from cannula by passing taila till the tip of cannula then it is carefully introduced into the uterus through the cervical canal, taila is pushed in uterine cavity very slowly with constant pressure for 2 to 3 minutes.

PASCHAT KARMA:

- The sterilised Yoni pichu kept in the vagina near cervix. The patient is asked to lie down for thirty minutes in lateral & head low position.
- The patient was observed for 1-2 hrs for any complaint of pain, bleeding, or discomfort, and for changes in the pulse, blood pressure, temperature etc.
- The pichu is retained for 2-3 Hour or till she gets urge of micturition, then it is removed by pulling the thread which is tied to the pichu, after sitting in squatting position. Patient is advised to take warm water bath, drink lukewarm water & laghu aahara sevana after evacuation of basti, neither talk loudly nor indulge in anger, not to travel in uneven roads, not to seat in uneven posture, to avoid day-sleep & intercourse.[79]

OBSERVATIONS & RESULTS

Table no.24: Frequency distribution of age of patients

	Age Group				
	Age Group	Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	21-25	7	23.33%	23%	23.33%
	25-30	10	33.33%	33%	56.67%
	31-35	11	36.67%	37%	93.33%
	35 and above	2	6.67%	7%	100%
	Total	30	100%	100%	

Graph 1: Showing statistic for age group

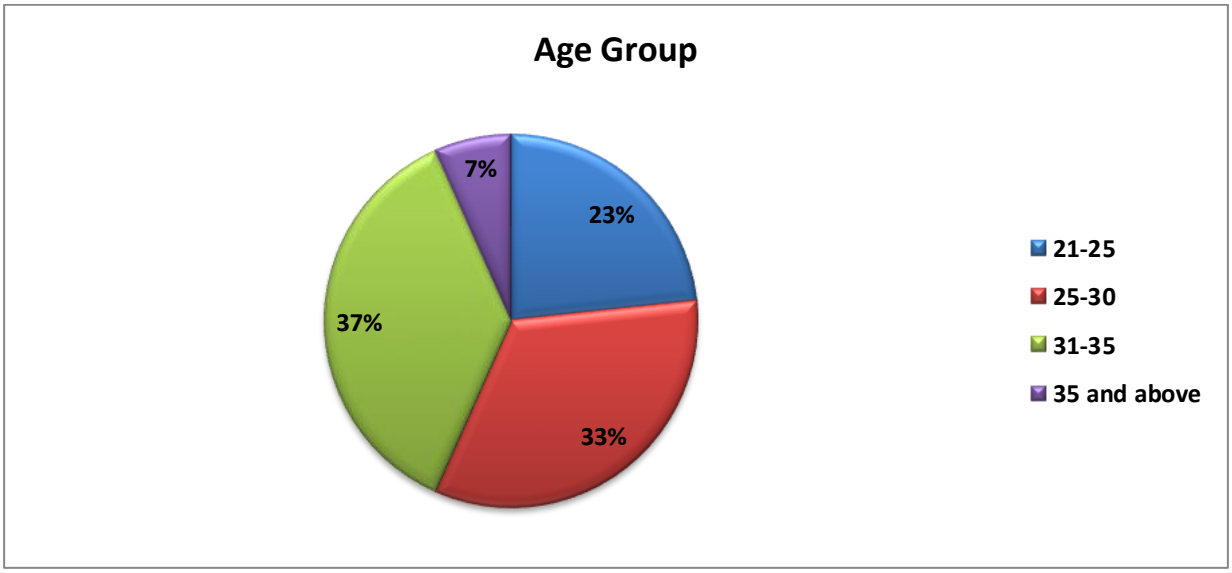


Table no.25: Frequency distribution of Marital Status

	Marital Status				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Married	30	100.00%	100%	100.00%
	Un Married	0	0.00%	0%	
	Total	30	100%	100%	

Graph 2: Showing statistic for Marital Status

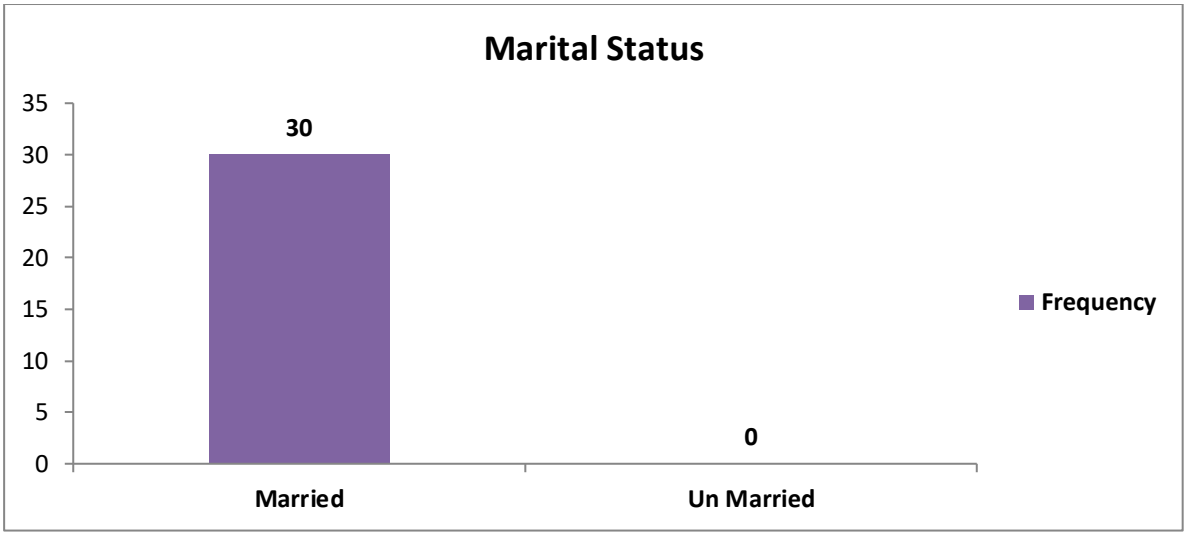


Table no.26: Frequency distribution of Parity

	Parity				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Parous	18	60.00%	60%	60.00%
	Nulliparous	12	40.00%	40%	100.00%
	Total	30	100%	100%	

Graph 3: Showing statistic for Parity

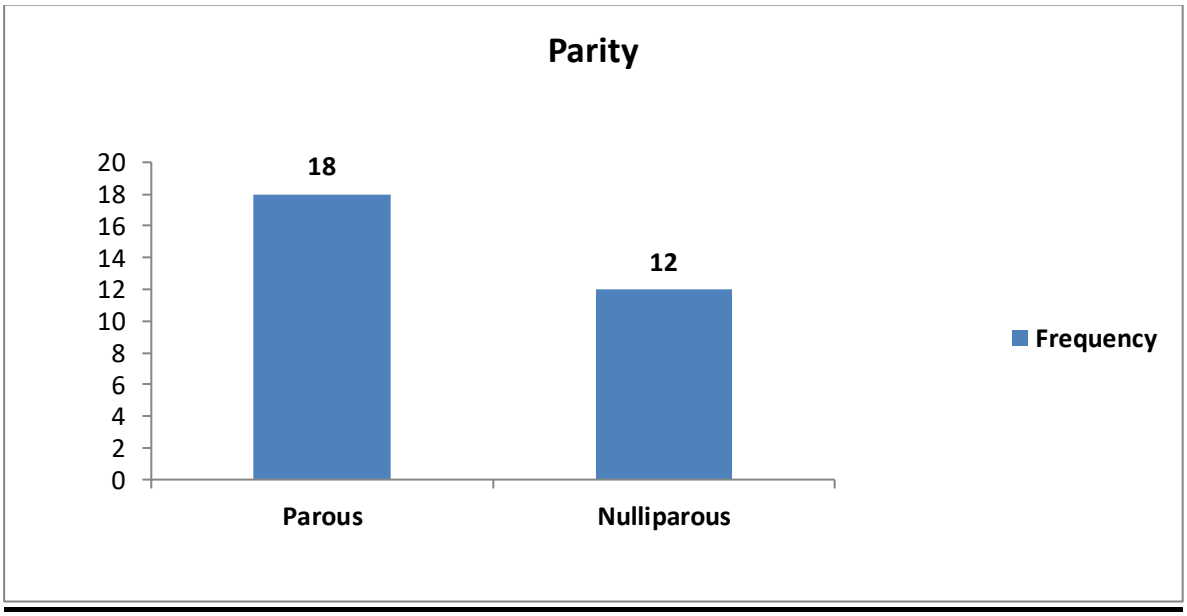


Table no.27: Frequency distribution of Socio Economic Status

	Socio Economic Status				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Middle Class	17	56.67%	57%	56.67%
	Upper Middle Class	13	43.33%	43%	100.00%
	Total	30	100%	100%	

Graph 4: Showing statistic for Socio Economic Status

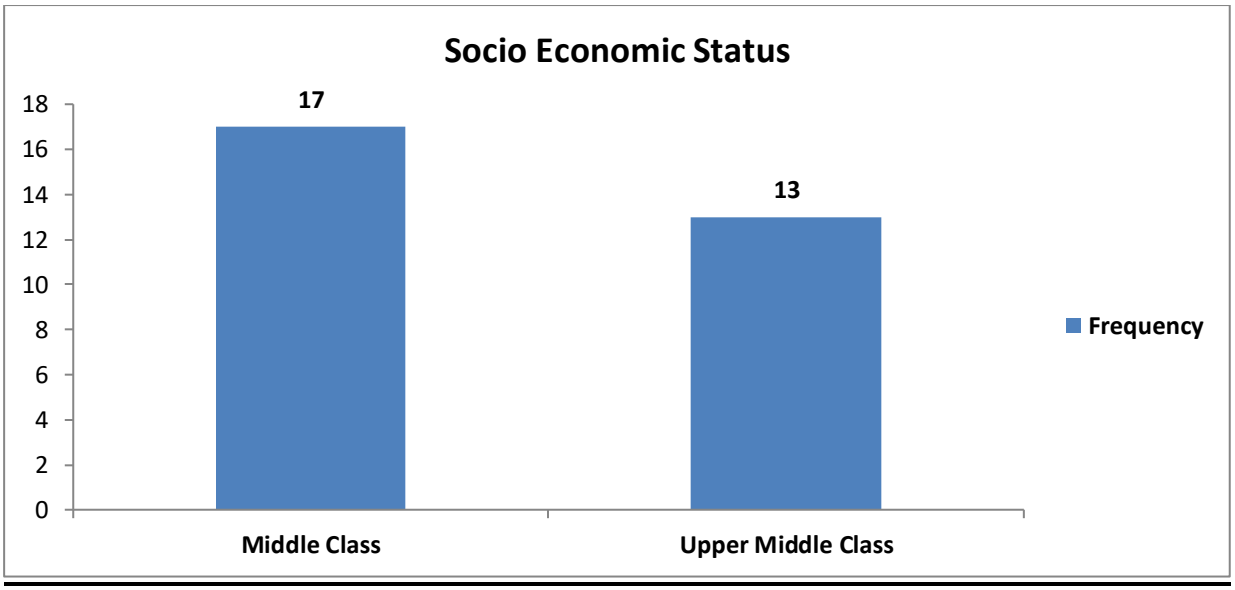


Table no.28: Frequency distribution of Religion

	Religion				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Hindu	25	83.33%	83%	83.33%
	Muslim	2	6.67%	7%	90.00%
	Christian	3	10.00%	10%	100.00%
	Total	30	100%	100%	

Graph 5: Showing statistic for Religion

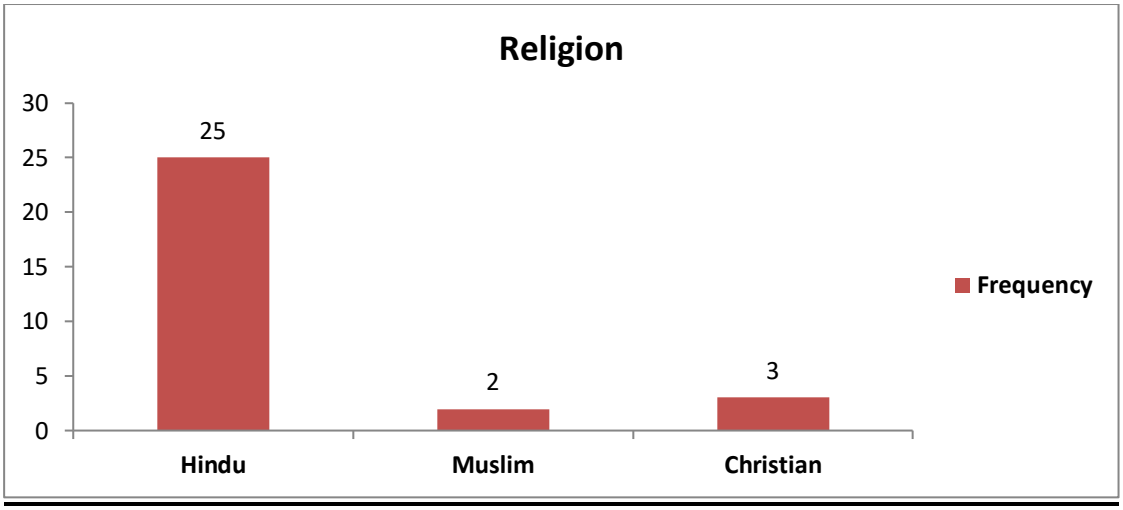


Table no.29: Frequency distribution of Occupation

	Occupation				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Engineer	4	13.33%	13%	13.33%
	House wife	23	76.67%	77%	90.00%
	Private Job	1	3.33%	3%	93.33%
	Teacher	2	6.67%	7%	100%
	Total	30	100%	100%	

Graph 6: Showing statistic for Occupation

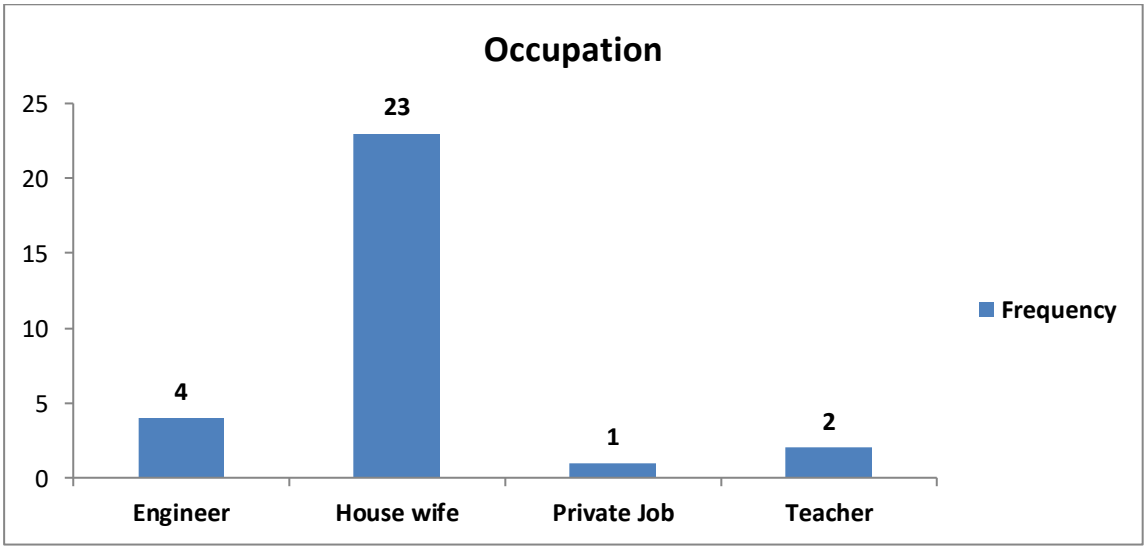


Table no.30: Frequency distribution of Quantity of menstrual flow Before Treatment

	Quantity of menstrual flow Before Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	2-3 Pads / Day (Normal)	0	0.00%	0%	0.00%
	1 Pad / day	6	20.00%	20%	20.00%
	Half Pad / Day	14	46.67%	47%	46.67%
	Spotting	10	33.33%	33%	100%
	Total	30	100%	100%	

Graph 7: Showing statistic for Quantity of menstrual flow Before Treatment

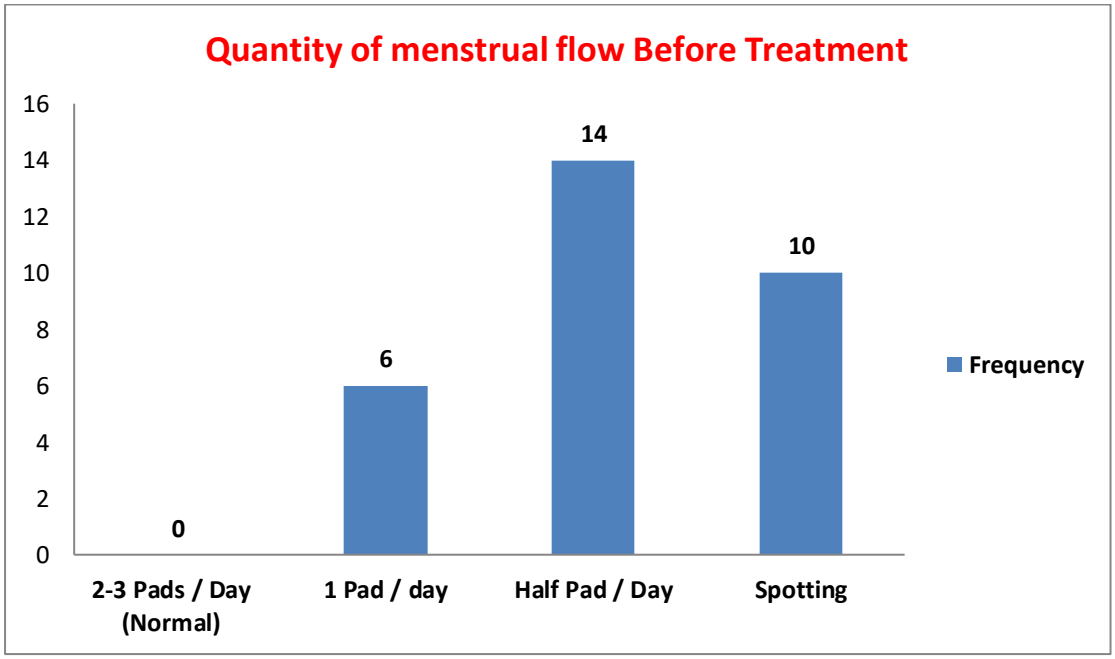
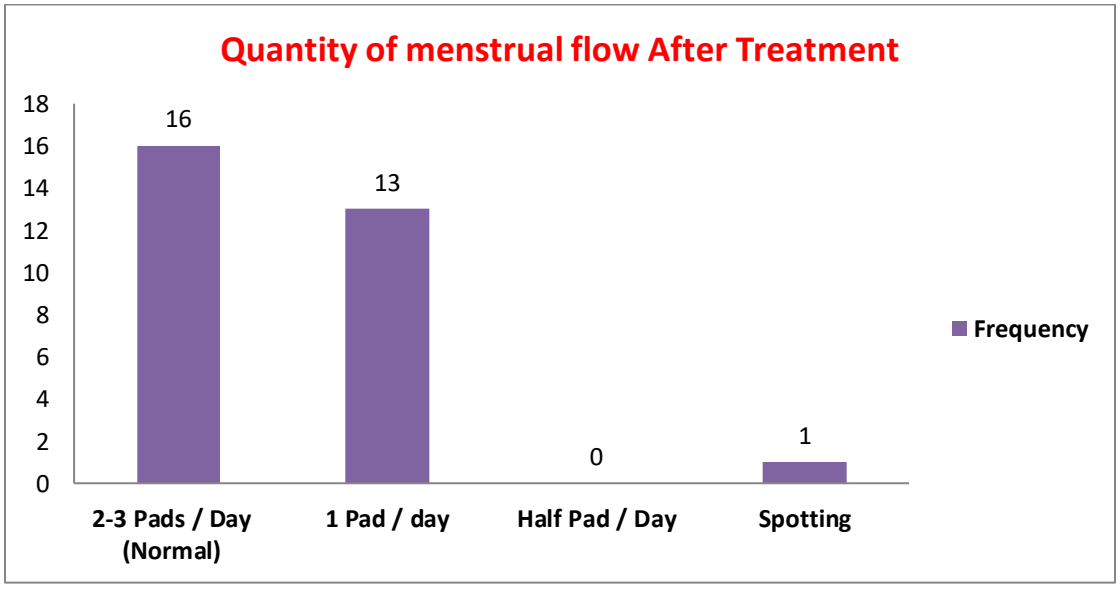


Table no.31: Frequency distribution of Quantity of menstrual flow After Treatment

	Quantity of menstrual flow After Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	2-3 Pads / Day (Normal)	16	53.33%	54%	53.33%
	1 Pad / day	13	43.33%	43%	96.67%
	Half Pad / Day	0	0.00%	0%	96.67%
	Spotting	1	3.33%	3%	100%
	Total	30	100%	100%	

Graph 8: Showing statistic for Quantity of menstrual flow After Treatment



Graph 9: Showing statistic for Quantity of menstrual flow Before and After Treatment

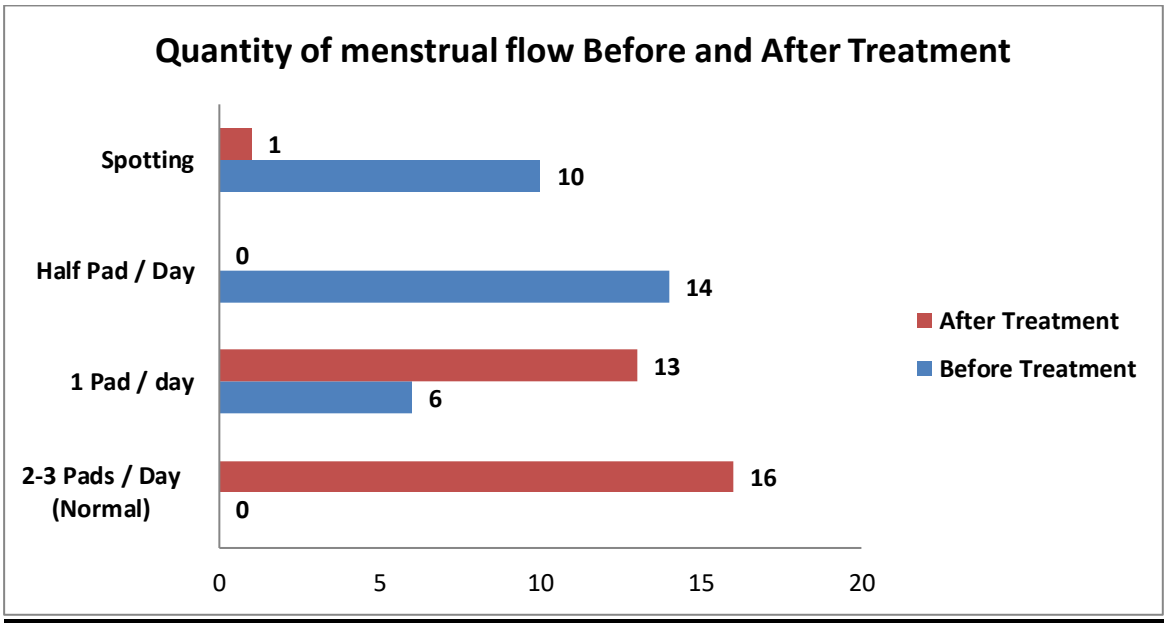


Table no.32: Frequency distribution of Duration of menstrual cycle Before Treatment

	Duration of menstrual cycle Before Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	4-7 Days (Normal)	0	0.00%	0%	0.00%
	3 Days	3	10.00%	10%	10.00%
	2 Days	17	56.67%	57%	56.67%
	1 Day	10	33.33%	33%	100%
	Total	30	100%	100%	

Graph 10: Showing statistic for Duration of menstrual cycle Before Treatment

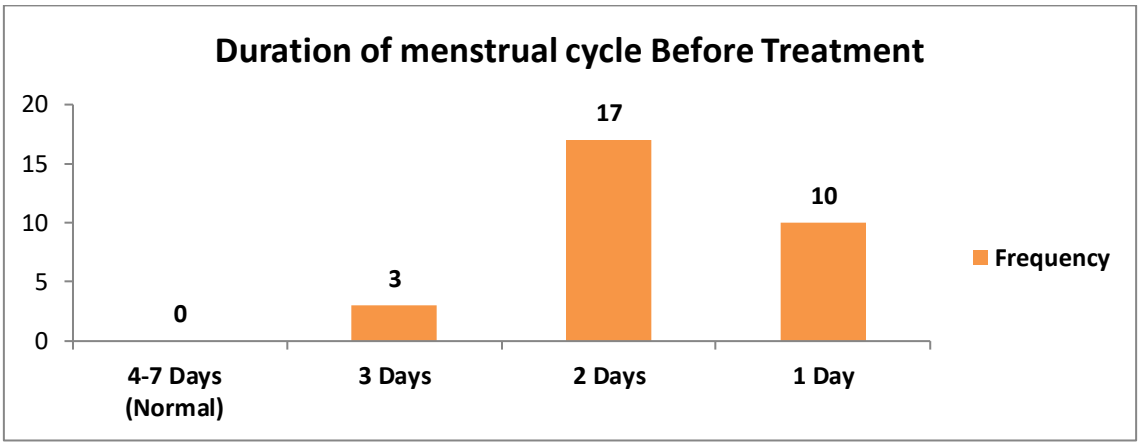
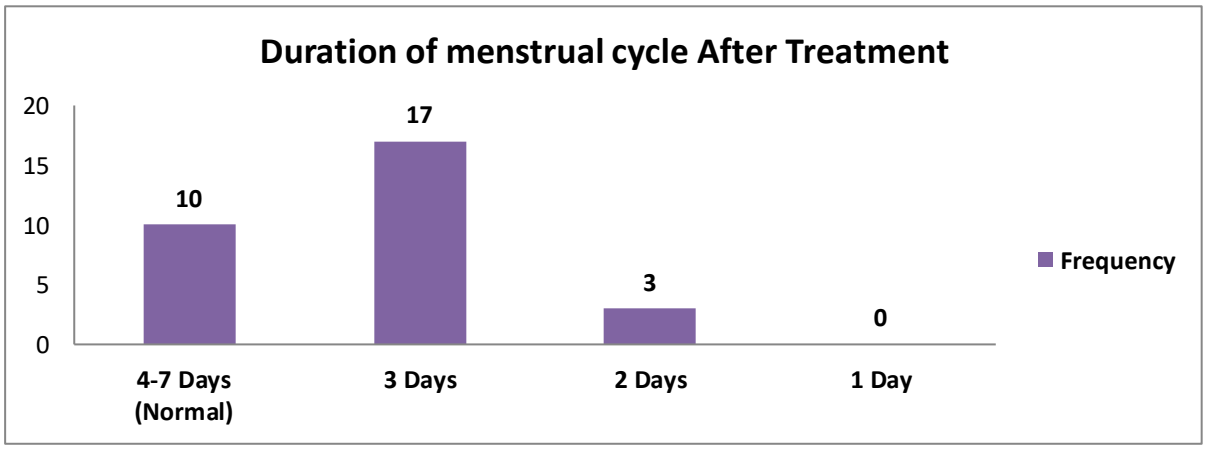


Table no.33: Frequency distribution of Duration of menstrual cycle After Treatment

	Duration of menstrual cycle After Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	4-7 Days (Normal)	10	33.33%	33%	33.33%
	3 Days	17	56.67%	57%	90.00%
	2 Days	3	10.00%	10%	100.00%
	1 Day	0	0.00%	0%	
	Total	30	100%	100%	

Graph 11: Showing statistic for Duration of menstrual cycle After Treatment



Graph 12: Showing statistic for Duration of menstrual cycle Before and After Treatment

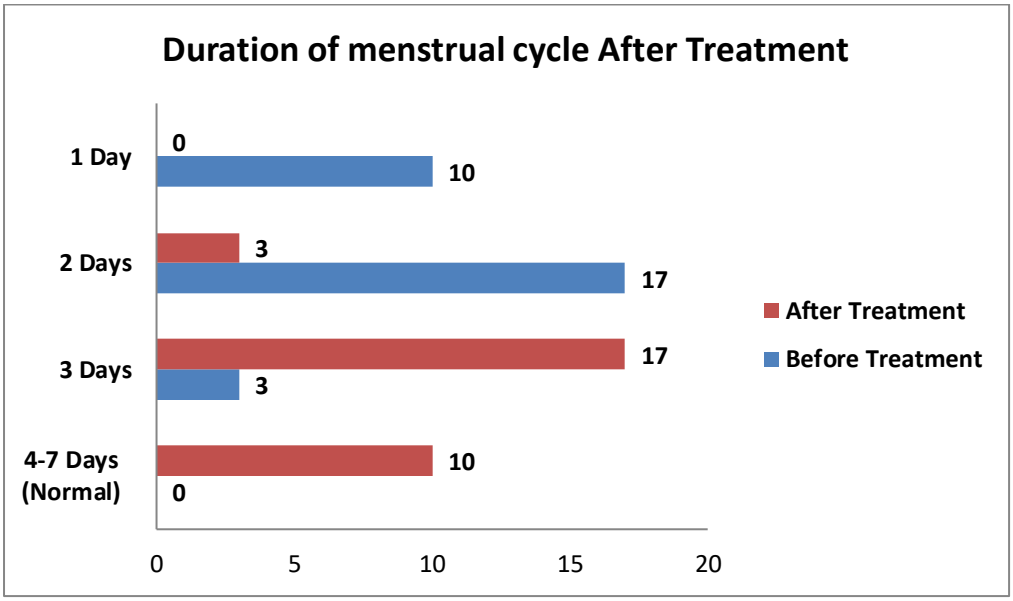


Table no.34: Frequency distribution of Pain in lower abdomen Before Treatment

	Pain in lower abdomen Before Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	No Pain (Normal)	1	3.33%	3%	3.33%
	Mild Pain	11	36.67%	37%	40.00%
	Moderate Pain	14	46.67%	47%	86.67%
	Severe Pain	4	13.33%	13%	100%
	Total	30	100%	100%	

Graph 13: Showing statistic for Pain in lower abdomen Before Treatment

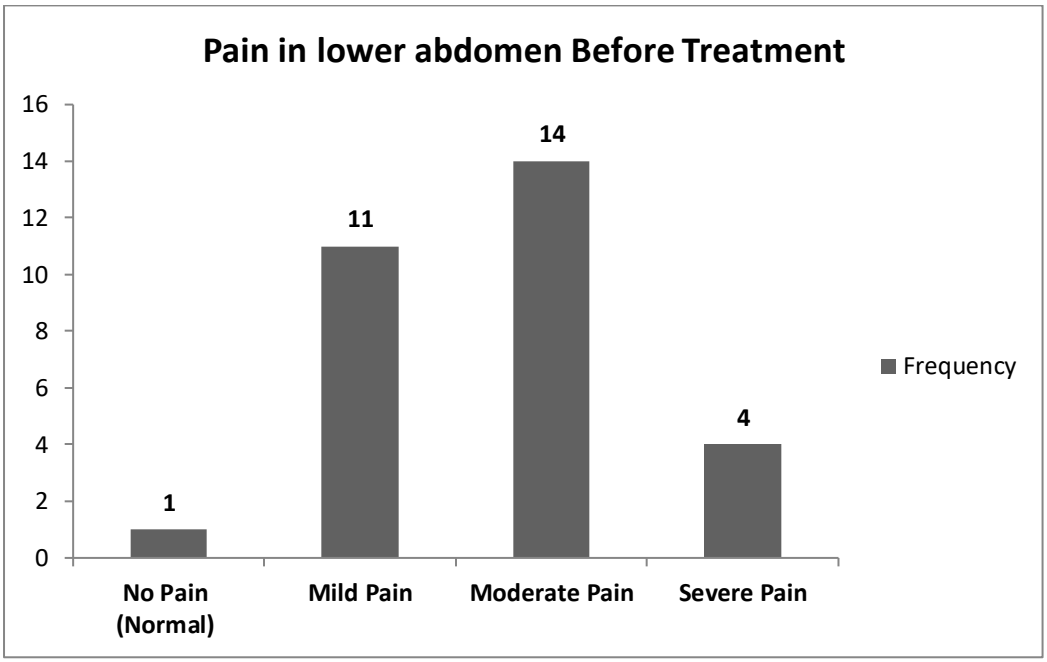
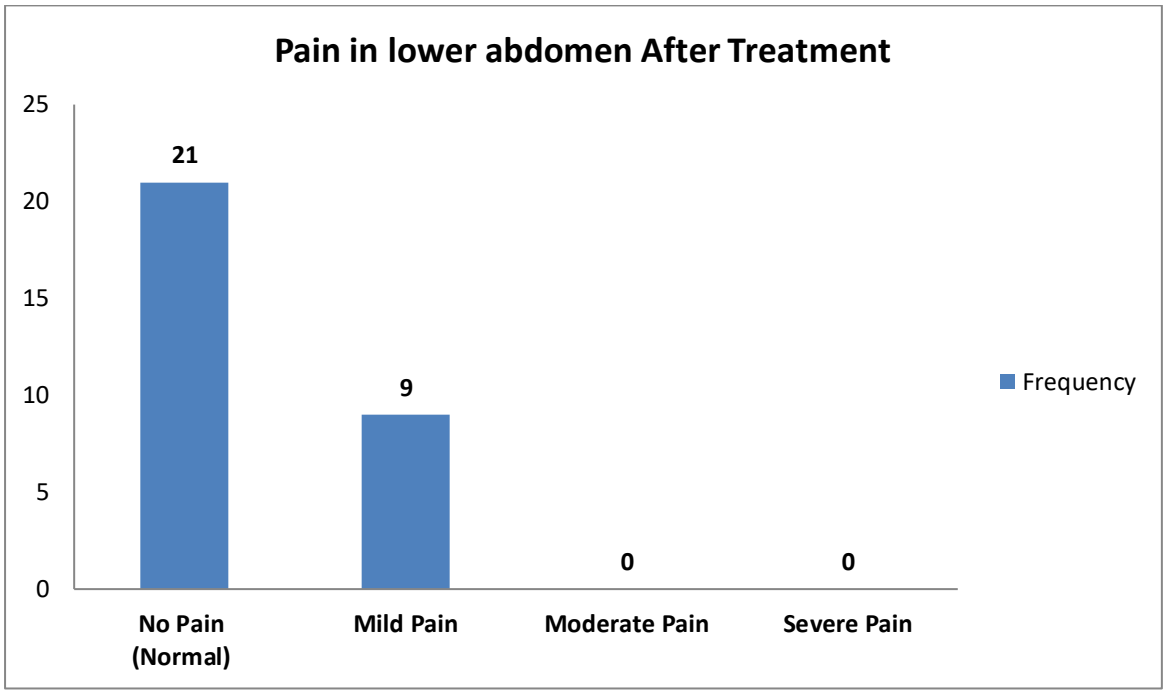


Table no.35: Frequency distribution of Pain in lower abdomen After Treatment

	Pain in lower abdomen After Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	No Pain (Normal)	21	70.00%	70%	70.00%
	Mild Pain	9	30.00%	30%	100.00%
	Moderate Pain	0	0.00%	0%	
	Severe Pain	0	0.00%	0%	
	Total	30	100%	100%	

Graph 14: Showing statistic for Pain in lower abdomen After Treatment



Graph 15: Showing statistic for Pain in lower abdomen Before and After Treatment

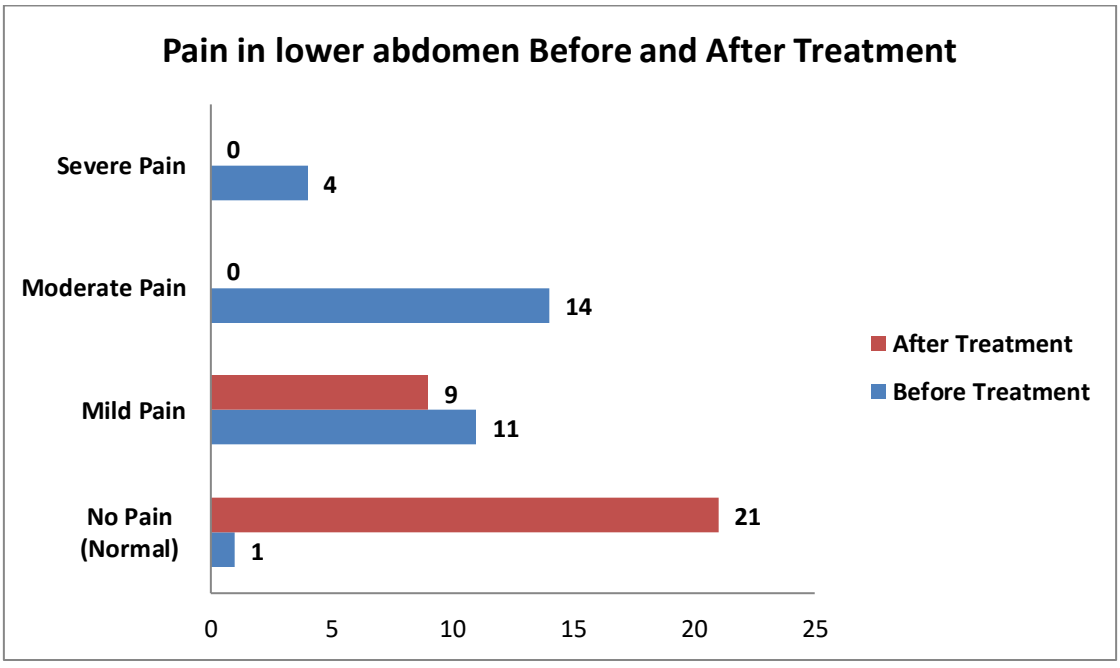


Table no.36: Frequency distribution of Endometrial Thickness Before Treatment

	Endometrial Thickness Before Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	10-12 mm (Normal)	7	23.33%	23%	23.33%
	13-15 mm	9	30.00%	30%	53.33%
	16-18 mm	12	40.00%	40%	93.33%
	>18 mm	2	6.67%	7%	100%
	Total	30	100%	100%	

Graph 16: Showing statistic for Endometrial Thickness Before Treatment

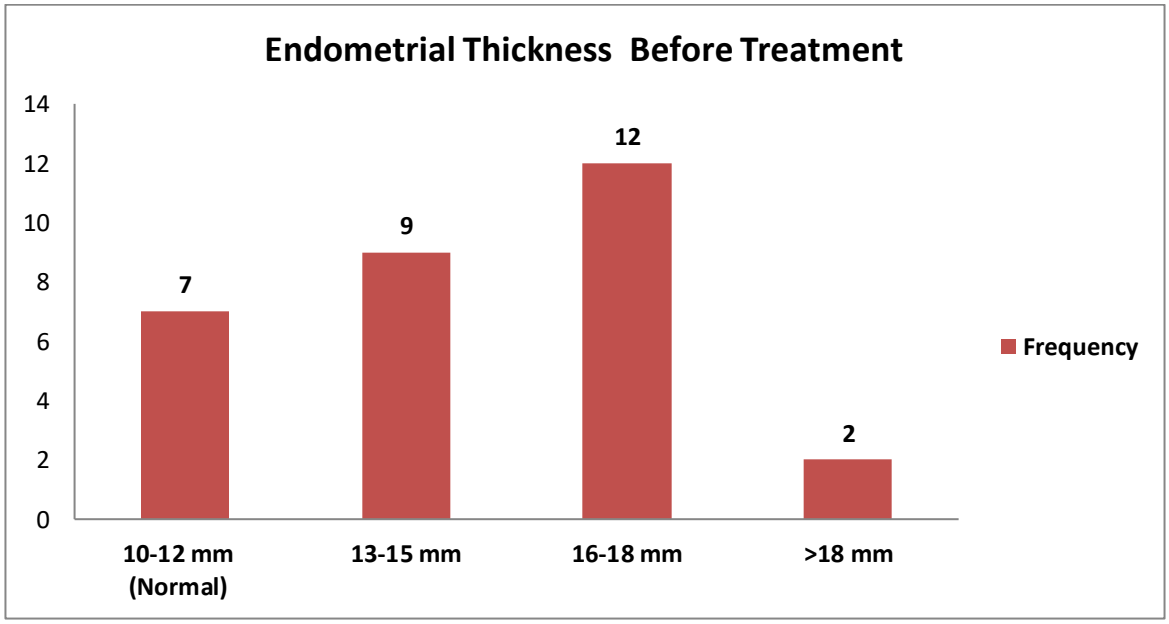
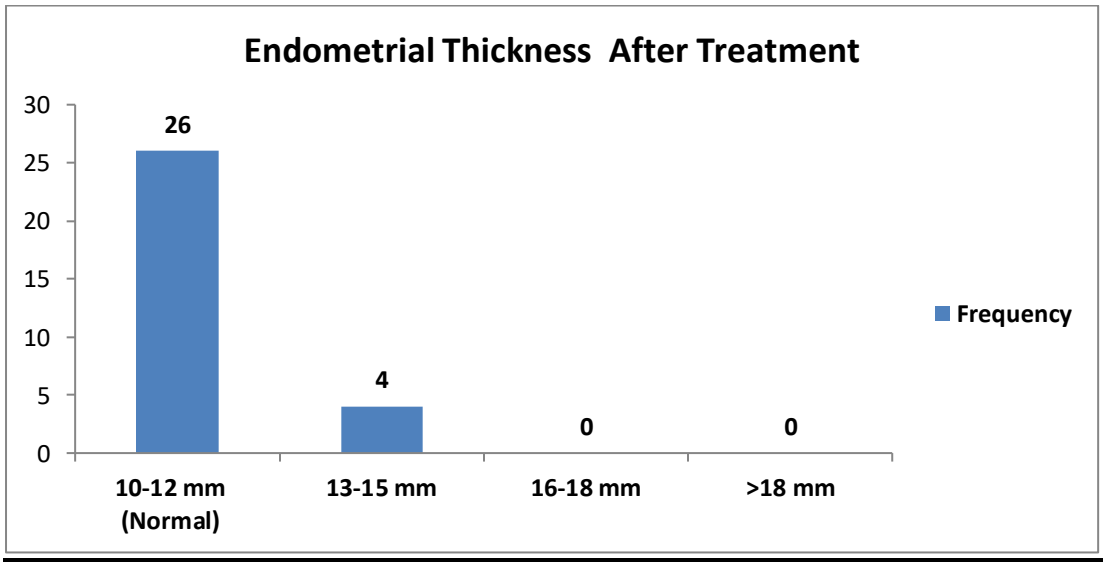


Table no.37: Frequency distribution of Endometrial Thickness After Treatment

	Endometrial Thickness After Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	10-12 mm (Normal)	26	86.67%	87%	86.67%
	13-15 mm	4	13.33%	13%	100.00%
	16-18 mm	0	0.00%	0%	
	>18 mm	0	0.00%	0%	
	Total	30	100%	100%	

Graph 17: Showing statistic for Endometrial Thickness After Treatment



Graph 18: Showing statistic for Endometrial Thickness Before and After Treatment

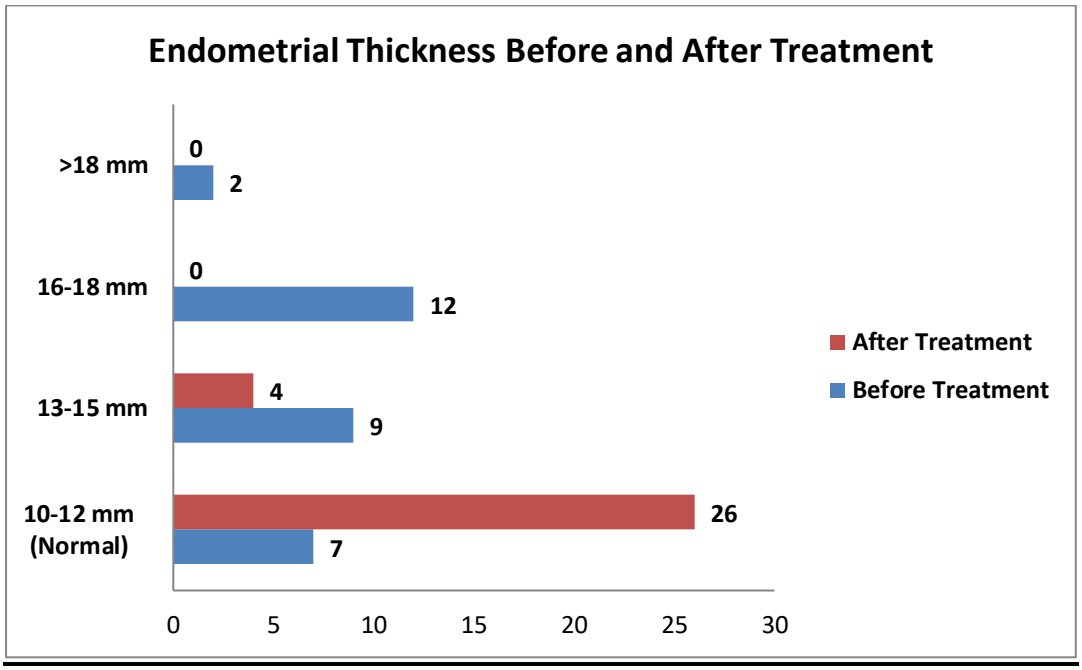


Table no.38: Frequency distribution of Levels of FSH & LH Before Treatment

	Levels of FSH & LH Before Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Normal (Normal)	9	30.00%	30%	30.00%
	Mild altered	9	30.00%	30%	60.00%
	Moderately altered	11	36.67%	37%	96.67%
	Excess altered	1	3.33%	3%	100%
	Total	30	100%	100%	

Graph 19: Showing statistic Levels of FSH & LH Before Treatment

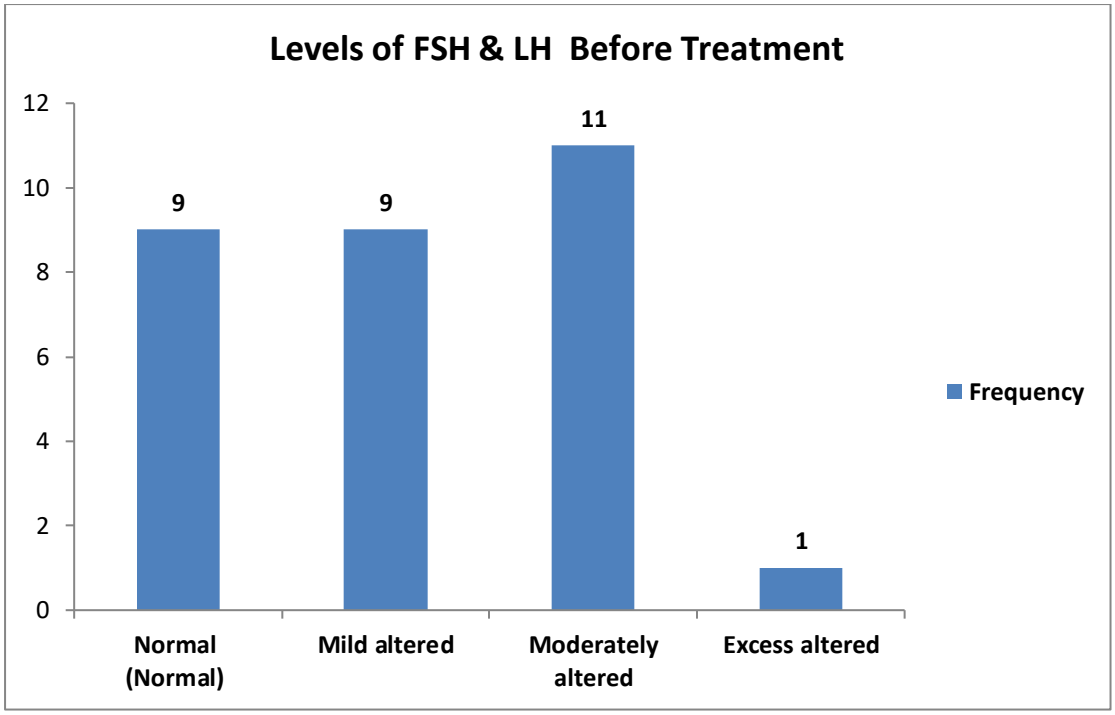
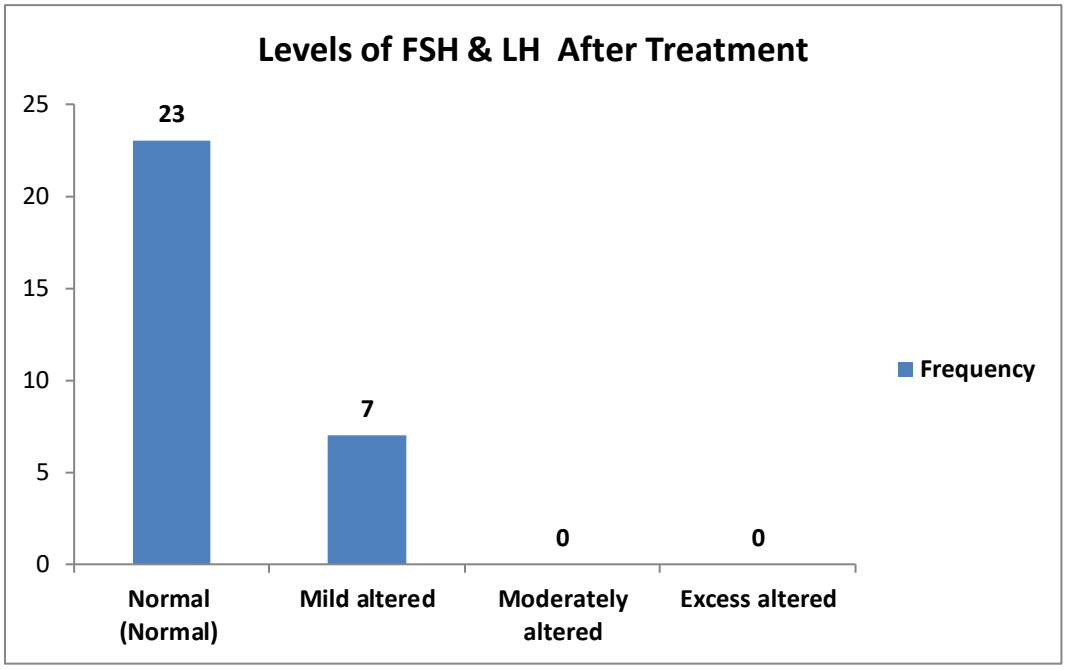


Table no.39: Frequency distribution of Levels of FSH & LH After Treatment

	Levels of FSH & LH After Treatment				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Normal (Normal)	23	76.67%	77%	76.67%
	Mild altered	7	23.33%	23%	100.00%
	Moderately altered	0	0.00%	0%	
	Excess altered	0	0.00%	0%	
	Total	30	100%	100%	

Graph -20: Showing statistic Levels of FSH & LH After Treatment



Graph -21: Showing statistic Levels of FSH & LH Before and After Treatment

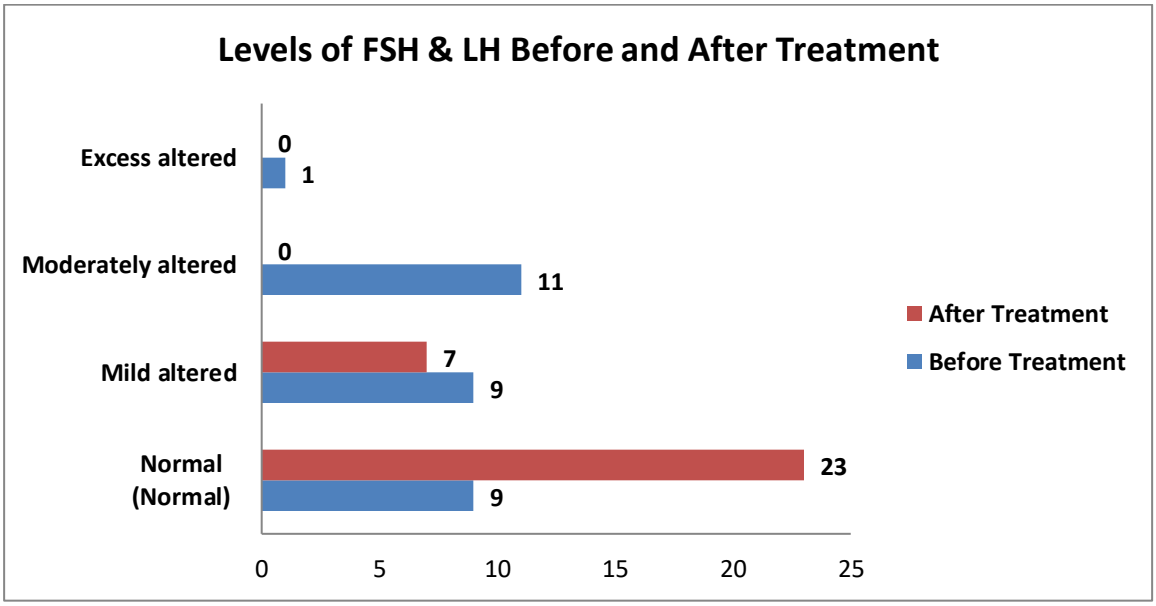


Table no.40: Frequency distribution of Overall Scoring

	Overall Scoring				
		Frequency	Percentage	Valid Percentage	Cumulative Percentage
Valid	Moderate improvement	12	40%	40%	40%
	Mild improvement	1	3.33%	3%	43.33%
	Good improvement	17	56.67%	57%	100.00%
	Total	30	100%	100%	

Graph -22: Showing statistic Levels of Overall Scoring

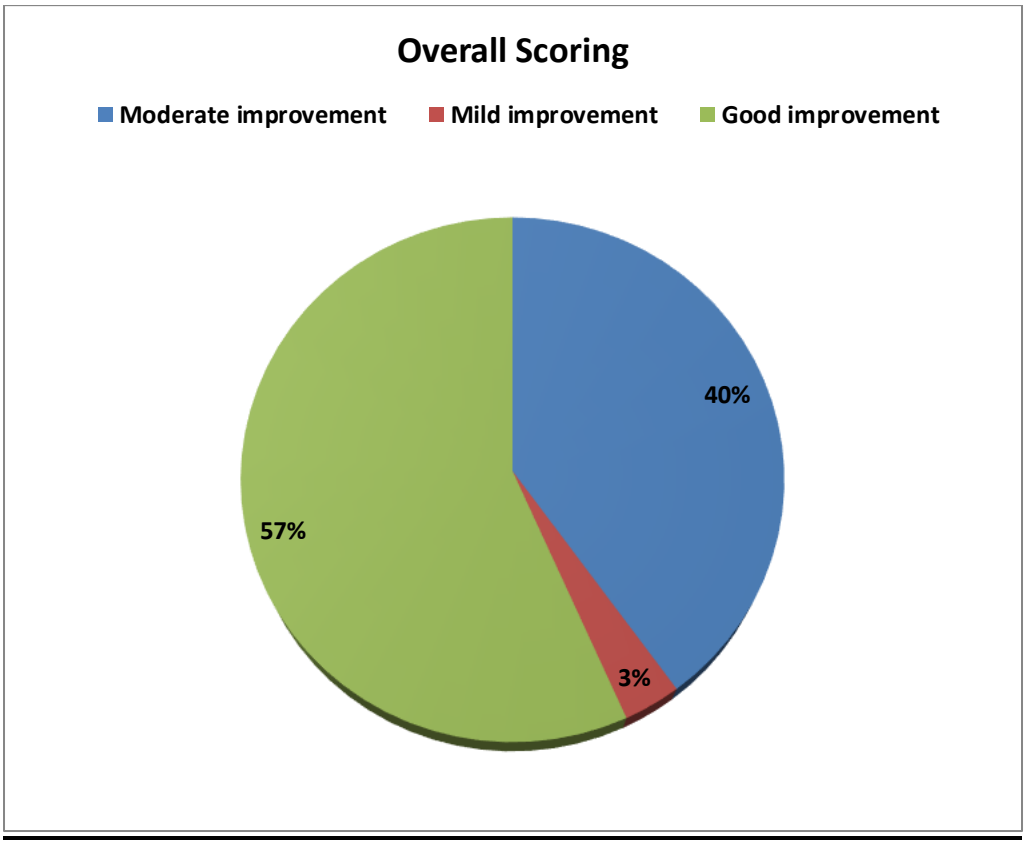


Table no.41: Paired Samples Statistics for the Group

Paired Samples Statistics for the Group					
		Mean	N	Standard Deviation	Standard Error
Pair 1	Quantity of menstrual flow Before Treatment	2.133333	30	0.730296743	0.1333333
	Quantity of menstrual flow After Treatment	0.533333	30	0.681445387	0.1244143
Pair 2	Duration of menstrual cycle Before Treatment	2.233333	30	0.626062316	0.1143028
	Duration of menstrual cycle After Treatment	0.766667	30	0.626062316	0.1143028
Pair 3	Pain in lower abdomen Before Treatment	1.7	30	0.749712589	0.1368782
	Pain in lower abdomen After Treatment	0.3	30	0.4660916	0.0850963
Pair 4	Endometrial Thickness Before Treatment	1.3	30	0.91538573	0.1671258
	Endometrial Thickness After Treatment	0.133333	30	0.345745904	0.0631243
Pair 5	Levels of FSH & LH Before Treatment	1.133333	30	0.89955289	0.1642351
	Levels of FSH & LH After Treatment	0.233333	30	0.430183067	0.0785403

Graph -23: Showing Paired Statistics for the clinical parameters

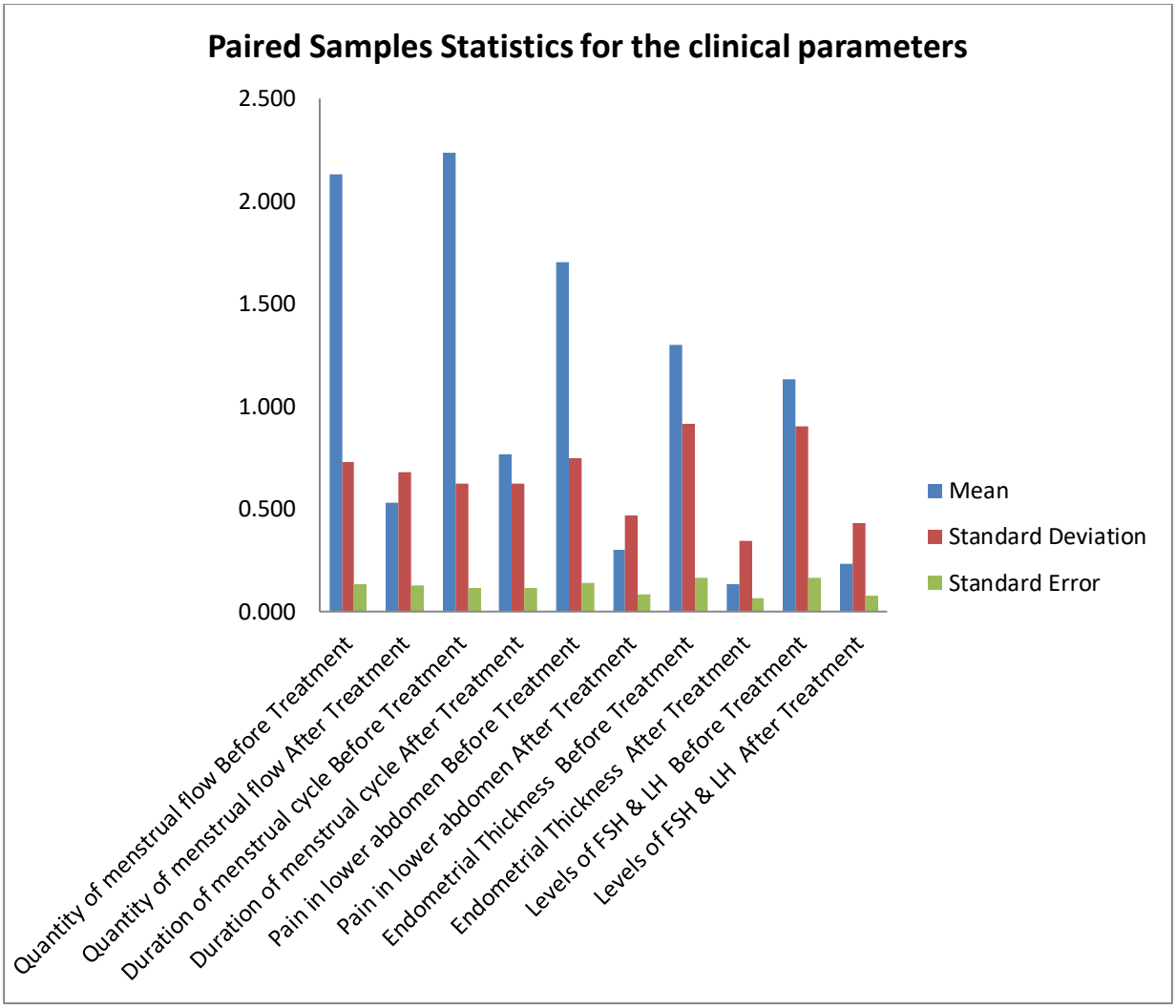


Table no.42: Paired Samples Statistics

	Mean Differences	Wilcoxon Sign rank Z	P Value	Remark
QUANTITY OF MENSTRUAL FLOW	1.6	4.63	< 0.01	Significant
DURATION OF MENSTRUAL CYCLE	1.47	4.55	< 0.01	Significant
PAIN IN LOWER ABDOMEN	1.4	4.87	< 0.01	Significant
ENDOMETRIAL THICKNESS	1.17	4.78	< 0.01	Significant
LEVELS OF FSH & LH	0.9	4.26	< 0.01	Significant

Table no.43: Showing rejection of Null hypothesis (H0) and acceptance of alternate (H1)

Null Hypothesis	Sig. Value	Result
H0 : There is no significant difference between Quantity of menstrual flow Before and After Treatment	0.00	Rejected
H0 : There is no significant difference between Duration of menstrual cycle Before and After Treatment	0.00	Rejected
H0 : There is no significant difference between Pain in lower abdomen Before and After Treatment	0.00	Rejected
H0 : There is no significant difference between Endometrial Thickness Before and After Treatment	0.00	Rejected
H0 : There is no significant difference between Levels of FSH & LH Before and After Treatment	0.00	Rejected

DISCUSSION

DISCUSSION

Discussion On Artavakshaya:

- Sushruta explained symptoms of Artava Kshaya as Yathochita Kaala Adarshanam (menstruation does not appear at its appropriate time or it is delayed or prolonged intermenstrual period), Alpa Artava Pravrutti (scanty menstruation), Yonivedana (vaginal or lower abdominal pain).
- In Artava Kshaya due to the vitiation of Tridosha i.e. Kapha Vata Vriddhi and Pitta Kshaya, Agni is in Mandavsatha at Koshtha and Dhatu level, Ama Utapatti, Rasa Dhatu Dushti, Kapha Avarana to Vata Dosha and Artava Vaha Srotorodha is present.

Process of Manifestation:

- In Artava Kshaya, Mithya Ahara- Vihara, Manasika Bhavas including stress and other etiological factors Tridosha as well as Agni vitiation ensues. Which results in augmentation and accumulation of Kapha Dosha and Vata Dosha.
- Due to impairment of Agni, improper digestion of victuals results in engenderment of Ama Annarasa which may further augment vitiated Kapha. Vitiated Rasa Dhatu and Rasagni Mandya cause Srotodusti.
- Progression of the pathological events is ensured by Uttarottara Dhatvagnimandhya and Uttarottara vitiation of Dhatus and Upadhatu of Rasa i.e., Artava get vitiated lead to Artava Dushti.
- Due to dushita kapha and vata, the poor proliferation causing inadequate surge of essential hormone like luteinizing hormone and due to poor performance of FSH, there is possible occurrence of thin endometrium, being insufficient for menstrual flow.
- Moreover, circulation of Ama Anna Rasa may further increase Srotorodha. Vitiated Rasa and augmented Kapha along with Vata Dosha provoke Dosha Dushya Sammurchna.
- Simultaneously Shukragni fail to perform their work results in destruction of male hormones results excessive androgen in body.

- In this Avastha, Medogata Lakshanas like Granthi etc. Vata Lakshanas like Angamarda, Agnimandya, Apravritti or Asamyaka Pravritti of Artava, Kaphaja Lakshana like Manda Pravritti or Apravritti of Artava, Sthoulya etc. may be observed.
- Organopathological changes start developing their various signs and symptoms in different system of the body i.e. oligomenorrhea, hypomenorrhea, polycystic ovaries, hirsutism, acne, hair fall etc. Doshadushya Sammurchna, if not treated leads to manifestation of Rupa of Vyadhi designated as Artava Kshaya.

Chikitsa Siddhanta Discussion

"Samprapti vighatanameva chikitsa" in artavakshaya, acc to Acharya Sushruta, the treatment is through administration of agneya dravyas, which corrects "artava pramanahani".

PROBABLE MODE OF ACTION OF ORAL DRUG LASHUNA KALPA[80]

- Due to Katu Rasa and Katu Vipaka, Ushna Virya, Teekshna and Pitta Rakta Vardhaka properties Lashuna is Aartava Janak. Thus, due to these special properties, Lashuna increases the menstrual blood flow.
- It's hot and penetrating qualities dilate decidual blood vessels and hence it enhances bleeding and acts as menstrual stimulator.
- Garlic is rich in flavonoids such as Quercetin and sulphur compounds, such as Allyl Propyl Disulphide that has been reported to be associated with decreased platelet aggregation and bleeding events.
- Smooth muscle relaxant Property: Ajoene and Alicin opens potassium ion channels causing a membrane hyper polarization and thus leads to a decrease of calcium ion inwards current into the vascular smooth muscle cell and finally induce vasodilation resulting from intracellular calcium decrease. Garlic-derived polysulfides stimulate the production of the vascular gasotransmitter hydrogen sulfide (H2S) and enhance the

regulation of endothelial nitric oxide (NO), which induce smooth muscle cell relaxation, vasodilation.

- Promoting favourable HDL/ LDL ratios: Due to its anti-hyperlipidaemic effect and favorable HDL/ LDL ratios, it is supposed to normalize the female steroidogenesis as Cholesterol is the main precursor of female reproductive hormones.

PROBABLE MODE OF ACTION OF RASNADI TAILA UTTARA BASTI:[78]

- The drugs used for preparing Rasnadi taila are all Anti-inflammatory & Anti-spasmodic in function- thereby relieves pain.
- Antioxidant property of drugs helps in proper maintenance of oestrogen production, ovulation and proper maintenance of luteal phase, progesterone- regulating whole H-P-O axis for proper physiology of menstruation.
- The drug used in Rasnadi Taila like Malati having Artavajanana property and Devadaru, Chitraka & Kantakari having Garbhashaya shodhana property which will help for increasing the flow of menstruation.
- The agneya dravyas when administered in form of taila locally in form of Uttaravasti helps in proper proliferation of endometrium and regulates the thickness of endometrium through its Snigdha guna.
- By prabhava, taila is vatahara and the asrayee bhoota dosha of pelvic region being vata gets pacified by taila prayoga.
- The drug Rasnadi taila when administered as Uttaravasti- the taila applied intrauterine and pichu placed intravaginal near posterior fornix get absorbed through process of resolution-diffusion into the cell membrane locally and distributed across the cervical region reaches the uterus through uterine first pass effect.
- The active principle gets diffused into local blood stream through pampiniform plexus, reaches till ovaries and thereby till hypothalamus through ductless circulation and through positive feedback mechanism.

- Godugdha used in Rasnadi taila having the property of Vatahara, Rasayana and Jeevaniya which will Correct the dushit apana vayu.
- Gomutra having Lekhaniya property which acts on endometrium by scraping the endometrium for increasing the menstrual blood flow.
- Anti-microbial & Anti-Fungal property of gomutra relieves infection and resumes proper cyclic phenomena of menstrual cycle.

Above all, the trial drug Lashuna kalpa and Rasnadi taila contains agneya dravyas which are by prabhava artava janaka and garbhashaya sankochaka properties, which will act on Artavakshaya.

- Theoretically, as per the text, the procedure Uttara basti administered is intended to act on garbhashaya directly. The virya of drug administered reach into the garbhashaya through following mechanisms.
  - Direct passive diffusion through local tissues.
  - Through cervical lumen
  - Through venous or lymphatic circulation.
  - Vascular exchange, diffusion between veins & arteries.

Uterine route as a mode of drug administration:

- Uterine Mucosa formed by a ciliated and secretory epithelium resting on a very cellular lamina propria.
- The number of ciliated cells and non-ciliated secretory cells varies along the oviduct. Secretory activity varies during the menstrual cycle.

Vascularity of endometrium in proliferative phase:

- Both the endometrial and sub endometrial vascularization index (VI) and vascularization flow index (VFI) increased during the proliferative phase, peaking approximately 3 days prior to ovulation before decreasing to 5 days post-ovulation.
- Thus, an oil-based drug when instilled in the uterus during the proliferative phase might be getting easily absorbed in the systemic circulation.
- The advantages of this route might be:

  i. Easiness of administration, Hepatic first pass-effect bypass.

ii. Low systemic drug exposure (namely in the case of products used for local conditions) and increased permeability for some drugs when comparing to the oral or other routes.

iii. Low enzymatic activity and the possibility of preferential transfer of absorbed drugs to the ovary (referred as the “first-ovary-pass effect”) may also benefit drug delivery

- Through perfusion, it is reported that drug applied in vaginal tissue reaches and acts on uterus within 5 hours of application.
- The size of particle below 90m TC is studied to be reaching uterus within a minute indicating direct transport mechanism involving aspiration through the cervical canal.
- The drug administered at the opening of the cervix travels towards uterus by the osmolarity of the Sneha dravya given. The Sneha dravya which remains inside may show systemic effect by being absorbed and transported into inferior vena cava by vaginal, retro sigmoidal, vesicular, and uterine veins.
- In recent time studies, it has been studied that any medicinal formulation placed intra vaginally, the drug absorbed in outer 1/3rd of the vagina passes uterus dramatically, influences first uterine pass effect and activating feedback mechanism till hypothalamus.
- This process occurs via following sequel.

Stimulating ovaries ⟶ pituitary stimulation ⟶ hypothalamus activation and thereby indicates the nourishment and rebounding functionality in the order vice versa i.e., Hypothalamus ⟶ pituitary gland ⟶ gonads (ovaries) ⟶ uterus (myometrium & endometrium) ⟶ till pacemakers of uterus.

Discussion On Clinical Study

The study was a randomised clinical study. A special proforma satisfying parameters of the study was prepared with all points of history taking, examination and investigations to procure the required data. The study was carried out with 30 patients with diagnosed artavakshaya, subjected to Lashuna Kalpa for 7 Days & Rasnadi taila Uttarabasti from D6- D10 of menstruation for 3 consecutive cycles. Follow up advised for next 3 cycles for observation.

Demographic description of patients

- In present study, out of total 30 patients 7 patients (23.33%) of the patients are of age group 21-25, 10 patients (33.33%) of them are from age group 25-30, 11 patients (36.67%) are of 31-35 age group and 2 patients (6.67%) are of 35 and above age group. The incidence in age 31-35 is high.
- Maximum of the patients are Hindus comprising of 25 (83.33%). Muslim patients are only 2 (6.67%) and Christian patient are 3 (10%) registered in the study.
- Out of 30 patient maximum number of patients 25 (83.33%) having mixed dietary habits and 5 (16.66%) are Vegetarians.
- Maximum patients registered in the study were housewives 23 (76.67%), Engineer 4 (13.33%), Private job 1 (3.33%), and Teacher 2 (6.67%).
- Maximum number of patients in the study are parous 18 (60%), and nulliparous are 12 (40%).
- Out of 30 patients Socio economic status of 13 (43.33%) are Uppermiddle class and 17 (56.67%) are Middle class.
- 100% of 30 patients are married, as the course of clinical intervention is per vaginal and is contra-indicated in virgins as per legal aspects, all the 30 patients with inclusive criteria are married.

DISCUSSION ON RESULTS

Quantity of menstrual flow

Out of 30 patients participated in the study mean score of the quantity of menstrual flow in no. of pads before treatment was 2.13 which reduced to 0.533333 after treatment with mean difference 1.667.

Statistical analysis shows, this difference is considered to be extremely statistically significant (p <0.01).

Duration of menstrual cycle

Out of 30 patients participated in the study mean score of the duration of menstrual cycle in no. of days before treatment was 2.233 which reduced to 0.7666 after treatment with mean difference 1.47. Statistical analysis shows, this difference is considered to be extremely statistically significant (p <0.01).

Pain in Lower Abdomen

Out of 30 patients participated in the study mean score of pain in lower abdomen before treatment was 1.7 which reduced to 0.3 after treatment with mean difference 1.4. Statistical analysis shows, this difference is considered to be extremely statistically significant (p <0.01).

Endometrial thickness

Out of 30 patients participated in the study mean score of endometrial thickness before treatment was 1.3 which reduced to 0.13 after treatment with mean difference 1.17. Statistical analysis shows, this difference is considered to be extremely statistically significant (p <0.01).

Levels of FSH & LH

Out of 30 patients participated in the study mean score of levels of FSH & LH before treatment was 1.13 which reduced to 0.23 after treatment with mean difference 0.9. Statistical analysis shows, this difference is considered to be extremely statistically significant (p <0.01).

Overall Result

Out of 30 patients 17 patients (57%) got good improvement, 12 patients (40%) got moderate improvement, 1 patient (3.33%) got mild improvement.

CONCLUSION & SUMMARY

CONCLUSION

Conclusion on study “A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA” is as follows.

Artavakshaya occurs due to imbalance of vata and kapha which leads to marga avarodha of artava vaha srotas and also occur by dhatukshaya janya. Artavakshaya is explained by Acharya Sushruta and is characterized by Yathochitakala - adarshanam, Alpartava and Yoni Vedana.

This condition could be compared to Hypomenorrhea based on its clinical presentation, also stated as precursor of Amenorrhea according to contemporary science.

The pathology includes altered or irregular H-P-O axis leading to infrequent and scanty menstruation. Assurance and hormonal therapy are accepted line of management.

According to Ayurveda, in artavakshaya the line of management is administration of agneya dravya which are artava vridhikara and in the form of taila which is vata hara.

As vata is the main causative factor for the disease manifestation, so it should be treated first. According to Ayurveda vitiation of Vata Dosha (mainly Apana Vayu) occurs due to its own prakopa (due to Vata vardhaka nidana), Dhatukshaya or Avarana. For Avrita Apana Vayu, treatment module should be agnideepaka, grahi, vatanulomana and pakvashaya shuddhikara. Lashuna fulfils all these properties thus has a beneficial effect in treatment of Oligo and Hypomenorrhea. Due to its platelet inhibitory, fibrinolytic, smooth muscle relaxant and HDL/LDL lowering actions, Lashuna (Allium sativum) increases the menstrual blood flow.

As Sthanika chikitsa is more efficacious in stree roga and uttarabasti being srestha among vata hara kriya yoga, the drug Rasnadi taila is given as uttaravasti.

The patients were given treatment for 3 months and observed for 1 yr. the clinical follow up inferred that cardinal symptom like scanty menstrual flow, prolonged intermenstrual period, lower abdominal pain markedly reduced.

The drugs in Lashuna Kalpa & Rasnadi taila proves “vriddhi samanaihi sarvesham” and tailam “vataharanam srestham”.

The administration of Rasnadi taila in the form of uttaravasti & Oral administration of Lashuna kalpa improved condition artavakshaya i.e menstrual flow increased in 96%, intermenstrual period regulated in 90% and lower abdominal pain and low backache relieved in 70%. The women between the age of 18 to 45 yrs, married, habituated to mixed diet are selected for study.

Out of 30 patients 17 patients (57%) got good improvement, 12 patients (40%) got moderate improvement, 1 patient (3.33%) got mild improvement.

Limitations of study:

The intervention being a vaginal procedure, administration in virgin women with similar pathology is the limitation of study.

Scope For Future Studies:

• Studies can be carried out on larger sample.

• Studies can be carried out in low AMH patients for good development of ovarian follicles.

SUMMARY

The present clinical study entitled “A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA” consisting of following parts.

1. Introduction
2. Aims and Objectives
3. Review of Literature
4. Drug Review
5. Materials and Methods
6. Observation and Results
7. Discussion
8. Conclusion & Summary

Introduction:

Brief information regarding Artavakshaya and hypomenorrhea, its incidence, ayurvedic line of management, importance of present study. It includes plan of the study and it is mainly the protocol of the present research work.

Aims and Objectives

Aims and objectives of study and need of the study are included.

Review of Literature:

At first Stree sareera rachana i.e., Anatomy of female reproductive system was explained briefly. The disease review elaborates the general description of Artavakshaya and its modern correlation – Hypomenorrhea. The etymological derivation, etiology, related anatomy, clinical manifestations, pathogenesis, prognosis and general principle of treatment of Artavakshaya is discussed here.

Drug review

The composition of Lashuna kalpa & Rasanadi Taila is detailed in the drug review. The properties of individual drug along with botanical description, chemical constituents, properties and indications are described.

Materials and methods:

The clinical plan of study, Preparation, and administration of the drug, criteria for selection of patient, parameters, assessment criteria, and investigations are described in the materials and methods.

Observation and Results:

The observations from the clinical study and results obtained from statistical data are represented in tabular forms and graphical representation and their description are mentioned.

Discussion:

This section deals with interpretation of the disease artavakshaya, mode of action of Lashuna kalpa & Rasnadi taila, observations obtained, and detailed analysis of result obtained.

Conclusion and summary:

The overall clinical study is concluded, and a brief study of the whole study is discussed in summary.

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LIST OF TABLES

LIST OF TABLES

Sl. No	Name of the Table	Page No
1.	Bones of Shroni	5
2.	Lakshana of Raja Srava	10
3.	Phase of menstruation and predominance of dosha	11
4.	Nidana of Artavakshaya	15
5.	Samprapti Ghataka	16
6.	Kalka Dravya of Rasnadi taila	30
7.	Lashuna	31
8.	Rasna	32
9.	Malati	33
10.	Guduchi	34
11.	Yashtimadhu	35
12.	Bala	36
13.	Kantakari	37
14.	Devadaru	38
15.	Chitraka	39
16.	Yuthika	40
17.	Tila taila	41
18.	Godugdha	42
19.	Gomutra	42
20.	Subjective Parameter Scoring	49
21.	Endometrial Thickness Score	49
22.	Scoring on FSH & LH	49
23.	Overall Results grading	49
24.	Frequency distribution of age of patients	55
25.	Frequency distribution of Marital Status	55
26.	Frequency distribution of Parity	56
27.	Frequency distribution of Socio Economic Status	57
28.	Frequency distribution of Religion	57
29.	Frequency distribution of Occupation	58
30.	Frequency distribution of Quantity of menstrual flow Before Treatment	59
31.	Frequency distribution of Quantity of menstrual flow After Treatment	60
32.	Frequency distribution of Duration of menstrual cycle Before Treatment	61
33.	Frequency distribution of Duration of menstrual cycle After Treatment	62
34.	Frequency distribution of Pain in lower abdomen Before Treatment	63

35.	Frequency distribution of Pain in lower abdomen After Treatment	64
36.	Frequency distribution of Endometrial Thickness Before Treatment	66
37.	Frequency distribution of Endometrial Thickness After Treatment	67
38.	Frequency distribution of Levels of FSH & LH Before Treatment	68
39.	Frequency distribution of Levels of FSH & LH After Treatment	69
40.	Frequency distribution of Overall Scoring	71
41.	Paired Samples Statistics for the Group	72
42.	Paired Samples Statistics	73
43.	Showing rejection of Null hypothesis (H0) and acceptance of alternate (H1)	74

List of Flowcharts

Sl. No	Name of Flowchart	Page No
1.	Samprapti of Artavakshaya	17

List of figures

Sl. No	Name of Figures	Page No
1.	Anatomical Structure of Uterus	20
2.	Anatomical Structure of Female genital tract	22
3.	Phases of Menstrual cycle	24
4.	Lashuna	31
5.	Rasna	32
6.	Malati	33
7.	Guduchi	34
8.	Yashtimadhu	35
9.	Bala	36
10.	Kantakari	37
11.	Devadaru	38
12.	Chitraka	39
13.	Yuthika	40
14.	Tila taila	41
15.	Godugdha	42
16.	Gomutra	42
17.	Lashuna kalpa(Churna)	43
18.	Rasnadi Taila Preparation	44
19.	Rasnadi Taila	44
20.	Uttar basti tray	52
21.	Garbhashayagata Uttarbasti	53

List of Graphs

Sl. No	Name of Graphs	Page No
1.	Showing statistic for age group	55
2.	Showing statistic for Marital Status	56
3.	Showing statistic for Parity	56
4.	Showing statistic for Socio Economic Status	57
5.	Showing statistic for Religion	58
6.	Showing statistic for Occupation	58
7.	Showing statistic for Quantity of menstrual flow Before Treatment	59
8.	Showing statistic for Quantity of menstrual flow After Treatment	60
9.	Showing statistic for Quantity of menstrual flow Before and After Treatment	61
10.	Showing statistic for Duration of menstrual cycle Before Treatment	62
11.	Showing statistic for Duration of menstrual cycle After Treatment	62
12.	Showing statistic for Duration of menstrual cycle Before and After Treatment	63
13.	Showing statistic for Pain in lower abdomen Before Treatment	64
14.	Showing statistic for Pain in lower abdomen After Treatment	65
15.	Showing statistic for Pain in lower abdomen Before and After Treatment	65
16.	Showing statistic for Endometrial Thickness Before Treatment	66
17.	Showing statistic for Endometrial Thickness After Treatment	67
18.	Showing statistic for Endometrial Thickness Before and After Treatment	68
19.	Showing statistic Levels of FSH & LH Before Treatment	69
20.	Showing statistic Levels of FSH & LH After Treatment	70
21.	Showing statistic Levels of FSH & LH Before and After Treatment	70
22.	Showing statistic Levels of Overall Scoring	71
23.	Showing Paired Statistics for the clinical parameters	73

ANNEXURES

DEPT. OF PRASUTI TANTRA & STREE ROGA

DR. NRS GOVT. AYURVEDIC COLLEGE, VIJAYAWADA

(AFFLIATED TO DR. NTR UNIVERSITY OF HEALTH SCIENCES, VIJAYAWADA)

TITLE OF THESIS: A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA

P. G SCHOLAR **:** DR. PATEL NIDHI SANJAYBHAI

GUIDE **:** DR. KOLA VIJAYA KUMARI, PROFESSOR & HOD, DEPT OF PRASUTI TANTRA & STREE ROGA, DR.NRS GOVT AYURVEDIC COLLEGE, VIJAYAWADA.

PLACE OF STUDY **:** DR. AL GOVT. AYURVEDIC HOSPITAL, VIJAYAWADA.

NATURE OF STUDY **:**

DATE OF COMMENCEMENT:

DATE OF COMPLETION **:**

PATIENT INFORMED CONSENT FORM

I, Sri/ Smt. / Kum _____

here by voluntarily give my free and whole hearted consent of participation in the clinical trial entitled "A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA."

The aims objectives and detailed methodology of the said study have been adequately explained to me, by the investigator to the fullest of my satisfaction and understanding. I have high regards for the professional capacity and ethical values of the investigation. I have understood that I may withdraw myself from the study at any point of time, whenever I wish.

Signature of Investigator **Signature of Patient/ Guardian**

Date:

Place:

THESIS CASE SHEET PROFORMA

Department of Prasuti Tantra and Stree Roga

Dr. NRS Government Ayurvedic College

Dr. AL Government Ayurvedic hospital

Vijayawada

Title: A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R. TO HYPOMENORRHEA

Research Scholar: Dr. Patel Nidhi Sanjaybhai	Guide: Dr. Kola Vijaya Kumari

Name	:	Case No	:
Age	:	OPD No	:
Occupation	:	Phone No	:
Address	:	Aadhar No	:
Husband name	:	Marital status	:
Occupation	:	Socio economic status	:

CASE HISTORY

1. **Chief Complaint (PradhanaVedana):**

2. **History of Present illness (Vedana Vruttanta):**

3. **History of Past illness (Poorva roga Vruttanta):**

4. Past Surgical History :

5. Family History :

6. Menstrual History

Age of menarche	:	LMP	:
Regularity of the periods	:	Cycle interval	:
Consistency	:	Duration	:
Quantity	:	Odour	:
Colour	:	Discharge	:
Sexual life	:	Contraception if any	:

7. Obstetrical History (Prasuti vrittantam):

G P L A D LCB

8. Vyaktigata Rupa (Personal History):

Education	:	Exercise	:
Hygiene	:	Community relations	:
Food habits	:	Addictions	:
Appetite	:	Bowel	:
Micturition	:	Sleep	:
Immunization	:	Consanguinity	:

9. Physical Examination :

General Examination: (A Part of Asta Sthana and anga- pratyanga pareeksha)

Stature	:	Height	:	Weight	:
B.P	:	R.R	:	P.R	:
Face	:	Neck	:	Upper limb	:
Lower limb	:	Chest	:	Complexion	:

Systemic examination:

Local examination: (Pradesika anga- pratyanga pareeksha)

1. Breast:

Lactation	:	Areola	:
Nipple	:	Axillae	:

2. Per abdomen:

Gynaecological examination:

a. Vulva/ External Genitalia :

b. Per speculum examination :

c. Per Vaginal examination :

10. Investigations:

1. CBP :

2. R.B.S. :

3. Thyroid Profile :

4. Hormonal Assay :

5. Viral Screening :

6. USG Abdomen :

Treatment plan :

- Lashuna Kalpa (Churna) 24 gm in two equal divided doses morning and evening for 7 days after menstruation.

  Anupana - Kshira or Ghrita or Navaneeta
- Uttar basti with Rasnadi Taila- After stoppage of bleeding or from 6th day of menstrual cycle for 5 days

Sl.No	Name of the parameter	1st month	2nd month	3rd month	Follow up
1	Bleeding amount in no of pads				
2	Duration of flow or menstrual period (In no of days)				
3	Pain in lower abdomen				

Assessment of study:

Sl.No	Name of the parameter	BT	AT
1	Bleeding amount in no of pads		
2	Duration of flow or menstrual period (In no of days)		
3	Pain in lower abdomen		
4	Endometrial Thickness		
5	Levels of FSH & LH		

Result:

Percentage improved:

Signature of PG Scholar Signature of Guide

MASTER CHART

S.No.	NAME	AGE	MARITAL STATUS	PARITY	SOCIOECONOMIC STATUS	DIET	RELIGION	OCCUPATION	QUANTITY OF MENSTRUAL FLOW		DURATION OF MENSTRUAL CYCLE		PAIN IN LOWER ABDOMEN		ENDOMETRIAL THICKNESS		LEVELS OF FSH & LH		IMPROVEMET %	OVERALL RESULT
									BT	AT	BT	AT	BT	AT	BT	AT	BT	AT		
1	G. Saundarya	33	Married	Parous	MC	Mixed	Hindu	House wife	2	0	3	1	2	0	2	0	0	0	88.8	Good
2	S. Susmitha	21	Married	Nulliparous	UMC	Mixed	Hindu	Engineer	3	0	2	0	2	0	2	0	2	0	100	Good
3	P. Pravina	33	Married	Parous	MC	Mixed	Hindu	House wife	2	1	2	1	1	0	1	0	1	0	71.4	Moderate
4	P. Lakshmi	24	Married	Parous	UMC	Mixed	Hindu	House wife	2	1	3	2	2	1	2	1	1	0	50	Moderate
5	Srilasya	25	Married	Nulliparous	UMC	Mixed	Hindu	Engineer	2	0	2	0	3	1	1	0	2	1	80	Good
6	G. Shubhalakshmi	34	Married	Parous	MC	Mixed	Hindu	House wife	1	0	3	2	2	0	2	1	1	0	66.6	Moderate
7	T. Adilakshmi	32	Married	Parous	MC	Veg	Hindu	House wife	2	1	3	1	2	0	1	0	2	1	70	Moderate
8	Nikhat Parveen	21	Married	Nulliparous	UMC	Mixed	Muslim	House wife	1	0	1	0	0	0	1	0	0	0	100	Good
9	B. Lavanya	26	Married	Nulliparous	MC	Mixed	Christian	House wife	1	0	2	0	2	1	2	1	0	0	71.42	Moderate
10	M. Sravani	33	Married	Nulliparous	MC	Mixed	Hindu	House wife	3	1	2	1	2	0	0	0	2	0	77.77	Good
11	S. Chaitanya	29	Married	Nulliparous	UMC	Mixed	Hindu	Teacher	3	3	3	2	1	0	0	0	3	1	40	Mild
12	J. Rani	27	Married	Nulliparous	MC	Mixed	Hindu	House wife	2	1	2	1	1	0	0	0	1	0	66.66	Moderate
13	T. Leelalakshmi	38	Married	Nulliparous	MC	Mixed	Hindu	House wife	1	0	2	1	1	0	1	0	1	1	66.66	Moderate
14	A. Prasanthi	24	Married	Nulliparous	MC	Mixed	Hindu	Engineer	2	0	1	0	2	1	0	0	0	0	80	Good
15	P. Anusha	28	Married	Parous	UMC	Mixed	Hindu	House wife	1	1	2	1	1	0	0	0	2	0	66.66	Moderate
16	Ch. Archana	22	Married	Parous	UMC	Mixed	Hindu	House wife	1	1	3	1	1	1	2	0	2	1	55.55	Moderate
17	K. Madhavilatha	32	Married	Parous	MC	Mixed	Hindu	House wife	2	1	2	0	1	0	2	0	1	0	87.5	Good
18	D. Sudharani	33	Married	Parous	MC	Mixed	Hindu	House wife	3	1	2	1	2	0	1	0	2	0	70	Moderate
19	SK. Fatehmunisa	34	Married	Nulliparous	MC	Mixed	Muslim	Teacher	3	0	3	1	2	0	2	0	1	0	90	Good
20	K.R. Shubhalakshmi	38	Married	Parous	UMC	Mixed	Hindu	House wife	2	0	1	1	3	0	3	1	0	0	77.77	Good
21	D. Bhavani	26	Married	Parous	UMC	Veg	Hindu	House wife	3	1	2	0	2	0	0	0	2	0	88.88	Good
22	L. Swarupa	26	Married	Parous	MC	Mixed	Christian	House wife	2	0	3	1	3	1	2	0	0	0	80	Good
23	Ch. Rajeshwari	30	Married	Parous	UMC	Mixed	Hindu	House wife	3	0	2	0	3	1	2	0	1	0	90	Good
24	A. Prasannalakshmi	22	Married	Nulliparous	MC	Veg	Hindu	Pvt. Job	3	1	2	0	2	1	0	0	2	1	66.66	Moderate
25	Shivachamundeshwari	29	Married	Parous	MC	Veg	Hindu	House wife	2	0	3	1	1	0	1	0	0	0	85.71	Good
26	G. Tejaswini	26	Married	Parous	UMC	Mixed	Hindu	House wife	2	0	2	0	1	0	2	0	0	0	100	Good
27	B. Janani	32	Married	Parous	MC	Mixed	Hindu	House wife	2	1	2	1	1	0	2	0	2	1	66.66	Moderate
28	N. Gautami	31	Married	Parous	UMC	Veg	Hindu	House wife	3	1	3	1	2	0	3	0	1	0	83.33	Good
29	P. Jyothi	30	Married	Parous	MC	Mixed	Christian	House wife	2	0	2	1	1	0	1	0	0	0	83.33	Good
30	B. Sindhuja	31	Married	Nulliparous	UMC	Mixed	Hindu	Engineer	3	0	2	1	2	1	1	0	2	0	80	Good

Dr.YSR UNIVERSITY OF HEALTH SCIENCES::A.P::VIJAYAWADA – 520 008

File No. 3301/A4/ MD(A)/Diss/21-22

Dt.24.04.2023

GOVT. AYURVEDIC COLLEGE INWARD NO. 606 INWARD DT. 28/4/2023 VIJAYAWADA-2

To
The Principal,
Dr.N.R.S. Government Ayurvedic College,
Bundar Road,
Vijayawada.

Sir,

Sub:- Dr. YSR UHS – Academic – M.D. (Ayurveda) Course 2021 -22 – Registration of Dissertation Topics – Allotment of Registration numbers – Reg.

Ref:- 1. Your Lr.2839/AC/2023, Dt.17-01-2023.
2.Admissions approved list.
3. Orders of the Vice-Chancellor, dt.24.04.2023.

28/4/2023

* * * * *

With reference to your letter cited above, it is to inform you that the dissertation topics of the following M.D. (Ayurveda) students for the year 2021-22 are registered with this University duly allotting the registration numbers as noted against their names and topics.

Sl. No	Candidate	Course	Topic	Reg. No
1	Dr.PATEL NIDHI SANJAYABHAI	Prasuti Tantra Evum Stree Roga	" A CLINICAL STUDY OF LASHUNA KALPA & RASNADI TAILA UTTAR BASTI IN ARTAVA KSHAYA W.S.R TO HYPOMENORRHEA".	A2110130301
2	Dr.NAGULA SINDHUJA	Prasuti Tantra Evum Stree Roga	" A CLINICAL STUDY OF KUTAJASTAKA VATI IN ASRGDARA W.S.R TO ABNORMAL UTERINE BLEEDING".	A2110130302
3	Dr.BUDDHAVARAPU SUSMITHA	Prasuti Tantra Evum Stree Roga	" A CLINICAL STUDY ON NASTARTAVA W.S.R. TO PCOD WITH PATHA LASUNADI VATI ORAL ADMINISTRATION WITH YOGA VASTI".	A2110130303
4	Dr.GADEPALLI LAKSHMIKAMAKSHI VADGEVI	Prasuti Tantra Evum Stree Roga	" A CLINICAL STUDY WITH DHATAKYADI TAILA UTTARA VASTI IN STREE VANDHYATVA W.S.R. TO TUBAL BLOCK".	A2110130304
5	Dr.MUCHAKALA SUNAYANA	Prasuti Tantra Evum Stree Roga	" A CLINICAL STUDY ON EFFECT OF PUMSAVANA VIDHI (NASYA) WITH KALYANAKA GHRITHA IN PUTRAGHNI YONIVYPATH W.S.R TO REPEATED ABORTIONS ALONG WITH ORAL ADMINISTRATION OF KALYANAKA GHRITHA".	A2110130305
6	Dr.KRUSHNA CHAITANYA PATNAIK	Prasuti Tantra Evum Stree Roga	" A COMPARATIVE STUDY OF VIRYAVARDHAKA CHURNA WITH AND WITHOUT VIRYAVARDHAKA BASTI IN KSHEENA SUKRA".	A2110130306
7	Dr.ASHIKAA K.S.	Kayachikitsa	" A CLINICAL STUDY TO COMPARE THE EFFECT OF MEDONIHANTHI YOGAM WITH SUDDHA GUGGULU VATI ON WEIGHT REDUCTION IN THE MANAGEMENT OF STHAULYA (OBESITY)".	A2110130201
8	Dr.CH. POLIKA NAGA KRISHNA SAI	Kayachikitsa	" A CLINICAL STUDY TO EVALUATE THE EFFICACY OF HUTABHUGADI CHOORNA IN THE MANAGEMENT OF PANDU ROGA (IRON DEFICIENCY ANAEMIA)".	A2110130202

Sl. No	Candidate	Course	Topic	Reg. No
9	Dr.KANKANTI MALATHI	Kayachikitsa	" A CLINICAL STUDY OF EVALUATE THE EFFICACY OF VIDANGAIDAGAJADI LEPA AND NAVAKA KASHAYA IN THE MANAGEMENT OF DADRU KUSTA – DERMATOPHYTOSIS (TINEA CORPORIS)".	A2110130203
10	Dr.B.N.V.S.PRAVALLIKA	Kayachikitsa	" A CLINICAL STUDY TO EVALUATE THE EFFECT OF BALA SAHACHARADI KASHAYA (INTERNALLY) AND KSHEERABALA TAILA (EXTERNALLY) IN THE MANAGEMENT OF VISWACHI W.S.R TO CERVICAL SPONDYLOSIS".	A2110130204
11	Dr.MOHAMMED JUNEED	Kayachikitsa	" A CLINICAL STUDY TO EVALUATE THE EFFICACY OF SUNTHY ADI, KWATHA CHURNA IN THE MANAGEMENT OF MUTRASHMARI (UROLITHIASIS)".	A2110130205
12	Dr.T.SOWJANYA	Kayachikitsa	" A CLINICAL STUDY TO EVALUATE THE EFFECT OF PANCHASAMA CHURNA IN THE MANAGEMENT OF AMAVATA W.S.R TO RHEUMATOID ARTHRITIS".	A2110130206
13	Dr.B.SWETHA	Kayachikitsa	" A CLINICAL STUDY TO EVALUATE THE EFFICACY OF PRASARINYADI KASHAYAM (INTERNALLY) & LAGHU VISHAGARBHA TAILAM (EXTERNALLY) IN THE MANAGEMENT OF KATIGRAHA W.S.R. TO LUMBAGO".	A2110130207
14	Dr.ALAKA THEJUS	Dravyaguna	" CLINICAL EVALUATION OF ANTIOBESITY (STHOULYAHARA) ACTIVITY OF ERYTHROXYLUM MONOGYNUM ROXB ANUKTADRAVYA".	A2110130401
15	Dr.AKULA PUSHPANJALI	Dravyaguna	" EVALUATION OF SWITRAHARA PROPERTY OF CLEISTANTHUS COLLINUS ROXB. (ANUKTADRAVYA)-A CLINICAL STUDY".	A2110130402
16	Dr.G.MEGHANA	Dravyaguna	" A CLINICAL EVALUATION OF MAMSAROHINI KANDA TWAK KASHAYA (SOYMIDA FEBRIFUGA A.JUSS) IN KASTARTAVA (DYSMENORRHEA)"	A2110130403
17	Dr.CHINNAMARAJU RAVALI	Dravyaguna	" EVALUATION OF PALASHA BEEJA (BUTEA MONOSPERMA LAN-KUTZE) EXTERNAL APPLICATION IN DADRU KUSHTA (DERMATOPHYTOSIS)- A CLINICAL STUDY.	A2110130404
18	Dr.VRUDDHULA VENKATA SANDHYA	Dravyaguna	" AN INVITRO STUDY OF VAYAHSTHAPANA DASEMANI HERBS ON VALI (WRINKLED SKIN)".	A2110130405

Sl. No	Candidate	Course	Topic	Reg. No
19	Dr.NEETU K.T	Rasa Shastra & Bhaishaya Kalpana	" PHARMACEUTICO –ANALYTICAL STUDY OF PARADADI MALAHARA AND ITS CLINICAL EVALUATION IN THE MANAGEMENT OF DUSHTAVRANA W.S.R. TO DIABETIC FOOT ULCER".	A2110130101
20	Dr.RAMYA BHARATHI P	Rasa Shastra & Bhaishaya Kalpana	" VALIDATION OF KESHYA PROPERTY OF SHILAJITH AND GUNJA EXTRACT (IN TRANSDERMAL PATCH FORM) IN KHALITYA W.S.R TO ANDROGEN ALOPECIA – A PHARMACEUTICO ANALYTICAL AND CLINICAL APPROACH".	A2110130102
21	Dr.N.S.L.SOWMYATHA	Rasa Shastra & Bhaishaya Kalpana	" PHARMACEUTICO – CLINICAL EVALUATION OF VICHARCHIKARI TAILA & RASA MANIKYA IN THE MANAGEMENT OF VICHARCHIKA – A COMPARATIVE STUDY".	A2110130103
22	Dr.RAVINDRA GNANASREE	Rasa Shastra & Bhaishaya Kalpana	" PHARMACEUTICO –ANALYTICAL STUDY OF VARNAKA GHRITA (EXTERNAL) AND MANJISHTA CHURNA (INTERNAL) AND ITS CLINICAL EVALUATION OF ITS VARNYA PROPERTY – A COMPARATIVE STUDY".	A2110130104
23	Dr.B.BHASKAR RAO	Rasa Shastra & Bhaishaya Kalpana	PHARMACEUTIVO – ANALYTICAL STUDY OF TAMRA SINDURA AND ITS CLINICAL EFFICACY IN THE MANAGEMENT OF KITIBA KUSTA (PSORIASIS)".	A2110130105

For further correspondence the registration number of the dissertation must be quoted without fail.

The Principal is further informed to make a note of the following:

a) Dissertation Topics must be registered with in six months from the date of joining the course.

b) Each topic of a student should be recommended by HOD and to be forwarded by Principal. In case of any repetition of the topics, the concerned Principal / HOD will be held responsible.

c) All dissertation topics in respect of **Admission Approved** students only should be sent to University for registration

Yours faithfully,
Sd/- REGISTRAR

// ATTESTED //

25/04/2023

JOINT REGISTRAR (ACADEMIC)

25/4/23

Copy to the individuals --- Through the Principal, **Dr.N.R.S. Government Ayurvedic College, Vijayawada.**

Copy to the Controller of Examinations, Dr. YSR UHS, Vijayawada.

GOVERNMENT OF ANDHRA PRADESH
DR NRS GOVERNMENT AYURVEDIC COLLEGE, VIJAYAWADA

Endorsement No 2839/Acad/2022 Dated 01/05/2023

//Copy Communicated//

7/5/2023
PRINCIPAL
PRINCIPAL
Dr. N.R.S. GOVT. AYURVEDIC COLLEGE
SUBMISSION ROAD, VIJAYAWADA-1

To

1) The HOD of Prasuti Tantra & Stree Roga,
   Dr NRS GAC, Vijayawada
2) The HOD of Kayachikitsa,
   Dr NRSGAC, Vijayawada
3) The HOD of Dravyaguna,
   Dr NRSGAC, Vijayawada
4) The HOD of Rasashastra & Bhaishjyakalpana,
   Dr NRSGAC, Vijayawada